Sidon Sets, Additive Energy, and the Optimal Cloning of Finite Phase-State Codebooks

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Abstract

Phase states $|\psi_x\rangle = d^{-1/2}\sum_j \omega^{xk_j}|j\rangle$, indexed by $x \in \mathbb{Z}_q$ and a key set $K = \{k_1, \ldots, k_d\} \subset \mathbb{Z}_q$, are the state family behind quantum fingerprinting and quantum hashing. We ask how well an adversary holding T copies of $|\psi_x\rangle$ can produce $T + k$ copies, and when a *finite* codebook is exactly as hard to clone as the continuous family of all phase states (the torus), whose optimal cloning fidelity was recently determined. The answer is governed by additive combinatorics of K. (i) The optimal $1 \to 2$ global cloning fidelity equals $(2d-1)/d^2$ if and only if K is a Sidon set. (ii) A B_{T+k} key set attains the torus optimum: analytically for $d = 2$ (all T) and for $T = 1$ (all d, k), and for $T = 2, 3$ (all d, k) by a proof whose finitely many computational steps are verified by standalone exact and interval-arithmetic checkers; we conjecture it for all T. (iii) For $T = 1$ we characterise exactly, for every k, the key sets that attain the torus optimum; the B_{1+k} condition is sufficient but not necessary. (iv) The optimal fidelity of fusing m phase states into the state of their sum equals $E_m(K)/d^{m+1}$, where E_m is the higher additive energy; Sidon sets minimise it for every m. Small Fourier bias, the standard design criterion for quantum hashing, does not control cloning. For finite orbits of non-abelian groups we prove a window theorem giving a dual-feasibility criterion, an obstruction, and a representation-theoretic selection rule for when matching the $(T + k)$-th moment suffices, and we give certified counterexamples for $SU(3)$ and $SU(2)$ showing that moment matching alone does not.

Keywords: quantum cloning, phase states, quantum hashing, Sidon sets, B_h sets, additive energy.
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1 Introduction

Quantum fingerprints [1] and quantum hash functions [2–5] map a classical string x to a short quantum state. In the phase form, $x \in \mathbb{Z}_q$ is encoded as

$$|\psi_x\rangle = \frac{1}{\sqrt{d}}\sum_{j=1}^{d} \omega^{xk_j}|j\rangle, \qquad \omega = e^{2\pi i/q}, \qquad K = \{k_1, \ldots, k_d\} \subset \mathbb{Z}_q. \tag{1}$$

The standard design criterion is that K be ε-biased, which bounds the overlaps $|\langle\psi_x|\psi_y\rangle|$ and hence collision resistance [3, 5]. When such states serve as quantum public keys or tokens [6], a second property matters: how well can an adversary holding T copies *counterfeit* $T + k$ copies?

For the continuous family of all phase states, $|\psi_\theta\rangle = d^{-1/2}\sum_j e^{i\theta_j}|j\rangle$ with θ uniform on the torus, the optimal $T \to T + k$ cloning fidelity is known: special cases go back to phase-covariant

cloning [7, 8], and the general value was recently obtained by Herasymenko, Lin, Verduyn Lunel and Grinko [9]. A finite codebook is a subset of the torus, so it can only be easier to clone. The question we answer is *when it is exactly as hard*.

Results.

- **Reduced dual (Lemma 3).** For every K and (T, k) the counterfeiting SDP reduces to a small convex program in the representation counts of K.
- **Sidon theorem (Theorem 7).** $P_K(1 \to 2) = (2d-1)/d^2$ if and only if K is Sidon.
- **Torus theorem (Theorem 9).** For a B_{T+k} key set, $P_K(T \to T+k) = P_{\mathrm{tor}}$ in the proved cases $d = 2$ (all T) and $T \le 3$ (all d, k); conjectured in general (Conjecture 14), with exact certificates on the finite boxes listed after the conjecture.
- **Fusion theorem (Theorem 26).** The optimal fusion fidelity is the additive energy $E_q(K)/d^3$.
- **Exact characterisation at $T = 1$ (Theorem 18).** For every k, $P_K(1, k) = P_{\mathrm{tor}}$ if and only if no additive class of K carries more weight than the torus maximum. B_{1+k} is sufficient but not necessary: Sidon sets that are not B_3 can attain the torus value (Section 5.4).
- **Multi-fold fusion (Theorem 28).** The optimal fidelity of fusing m phase states is the higher energy $E_m(K)/d^{m+1}$; Sidon sets minimise it for every m.
- **Negative results.** Small bias does not imply counterfeit resistance; the Sidon sets in our tables that are not B_3 fail at $T + k = 3$, and arithmetic progressions are worst in most cases tested (Section 5.3).
- **Non-abelian orbits (Section 7).** A window theorem for finite orbits $X = G \cdot f \subset H \cdot f$, and certified counterexamples ($SU(3)$ on $\mathrm{Sym}^2 \mathbb{C}^3$; $SU(2)$ spin 1) to the naive moment-matching principle.

The torus values themselves are not new: they are given in full by [9], with special cases in [7, 8]. What is new is the exact characterisation of the *finite* codebooks that attain them, the additive-energy form of optimal fusion, and the failure of the corresponding moment-matching principle for finite orbits of non-abelian groups. Section 8 discusses prior work in detail.

2 Setting and notation

Throughout, $K = (k_1, \ldots, k_d)$ consists of $d \ge 2$ distinct elements of $\mathbb{Z}_q$, and $x \in \mathbb{Z}_q$ is uniform.

Definition 1 (Counterfeiting value)**.** For $T \ge 1$, $k \ge 1$,

$$P_K(T, k) = \max_{\Phi} \ \mathbb{E}_{x \in \mathbb{Z}_q} \langle \psi_x^{\otimes(T+k)} | \, \Phi\big(\psi_x^{\otimes T}\big) \, | \psi_x^{\otimes(T+k)} \rangle,$$

the maximum over quantum channels (CPTP maps) Φ from $\mathrm{Sym}^T(\mathbb{C}^d)$ to $\mathrm{Sym}^{T+k}(\mathbb{C}^d)$; this is the average *global* fidelity with the ideal output $\psi_x^{\otimes(T+k)}$. The torus value $P_{\mathrm{tor}}(d; T, k)$ is the same quantity for $|\psi_\theta\rangle = d^{-1/2} \sum_j e^{i\theta_j} |j\rangle$ with θ uniform on $[0, 2\pi)^d$. Twirling any channel over the phase group $\theta \mapsto \theta + \phi$ preserves the average fidelity and produces a covariant channel, whose fidelity is the same on every phase state; hence the optimal average equals the optimal worst-case fidelity $F_{\mathrm{ph}}^\star(d, T, k)$ of [9, Thm. II.3; formal statement Thm. X.1, Eq. (16)].

Definition 2 (B_h sets, representation counts, additive energy)**.** K is a B_h set (mod q) if any two distinct *multisets* of size h drawn from K (repetition allowed, order ignored) have distinct sums mod q; B_2 sets are *Sidon* sets [10, 11]. A B_h set is $B_{h'}$ for every $h' \le h$ (add the same $h - h'$ elements to both multisets). By contrast, the counts r_m below are over *ordered* tuples. The ordered representation count is $r_m(v) = \#\{(j_1, \dots, j_m) \in [d]^m : \sum_i k_{j_i} \equiv v\}$, and the additive energy is $E_q(K) = \#\{(a, b, c, e) \in K^4 : a + e \equiv b + c \pmod q\}$.

We write $D_m = \binom{d+m-1}{m} = \dim \mathrm{Sym}^m(\mathbb{C}^d)$ and $W(d, T, k) = D_T / D_{T+k}$, Werner's optimal universal cloning fidelity [12]. For multi-indices $\gamma \in \mathbb{Z}^d_{\ge 0}$, $\mathrm{mult}(\gamma) = |\gamma|! / \prod_i \gamma_i!$.

3 The reduced dual

The counterfeiting SDP has dual $P_K(T, k) = \min\{\mathrm{tr}\, Y : Y \otimes I \succeq X\}$, with $X = \mathbb{E}_x \bar{\psi}_x^{\otimes T} \otimes \psi_x^{\otimes (T+k)}$ restricted to $\mathrm{Sym}^T \otimes \mathrm{Sym}^{T+k}$.

Lemma 3 (Reduced dual)**.** *For every K and (T, k),*

$$P_K(T,k) = \min_{z \in (0,\infty)^{\mathrm{supp}\, r_T}} \sum_{w \in \mathrm{supp}\, r_T} \frac{r_T(w)}{z_w} \quad s.t. \quad d^{-(2T+k)} \sum_w r_{T+k}(s+w)\, z_w \le 1 \qquad \forall s \in \mathbb{Z}_q,$$

the sums running over $w \in \mathrm{supp}\, r_T$; the minimum is attained.

Proof sketch; full proof in Appendix B. In occupation bases $|\alpha\rangle$ ($|\alpha| = T$) and $|\beta\rangle$ ($|\beta| = T+k$), with $c_\gamma = \sqrt{\mathrm{mult}(\gamma)}$, averaging over x gives $X = \sum_{s \in \mathbb{Z}_q} |w_s\rangle\langle w_s|$ with $w_s = d^{-(2T+k)/2} \sum_{K \cdot (\beta - \alpha) \equiv s} c_\alpha c_\beta |\alpha, \beta\rangle$. X and the constraint are invariant under $\bar{U}_x^{\otimes T} \otimes U_x^{\otimes (T+k)}$, $U_x = \mathrm{diag}(\omega^{x k_j})$, so an optimal Y may be taken block diagonal over the classes $\{\alpha : K \cdot \alpha \equiv w\}$. Within sector s, $Y \otimes I \succeq X$ holds iff $\sum_{(\beta, w)} c_\beta^2 d^{-(2T+k)} v_w^\dagger Y_w^{-1} v_w \le 1$ with $v_w(\alpha) = c_\alpha$. For fixed $g_w = v_w^\dagger Y_w^{-1} v_w$, Cauchy–Schwarz gives $\mathrm{tr}\, Y_w \ge \|v_w\|^2 / g_w = r_T(w)/g_w$, attained by a rank-one Y_w; put $z_w = g_w$ and use $\sum_{K \cdot \beta \equiv v} c_\beta^2 = r_{T+k}(v)$. Strong duality holds by Slater. □

Lemma 4. *For every K, $P_K(T, k) \ge P_{\mathrm{tor}}(d; T, k)$.*

Proof. The optimal torus cloner of [9] is covariant and achieves fidelity P_{tor} on every phase state, in particular on every $|\psi_x\rangle$. □

Lemma 5 (Moment bound)**.** *For any finite ensemble $\{\psi_x\}$, $P \le \dim\big(\mathrm{span}\{\psi_x^{\otimes T}\}\big)\, \lambda_{\max}\big(\mathbb{E}_x \psi_x^{\otimes (T+k)}\big)$.*

Proof. $\bar{\psi}^{\otimes T} \preceq \Pi_V$ on the span V of the inputs, so $X \preceq \Pi_V \otimes M$ with $M = \mathbb{E}\psi^{\otimes (T+k)}$, and $\mathrm{tr}(JX) \le \lambda_{\max}(M)\, \mathrm{tr}\big((\Pi_V \otimes I) J\big) = \lambda_{\max}(M) \dim V$ for every Choi operator J with $\mathrm{tr}_{\mathrm{out}}\, J = I$. □

Proposition 6 (Near-optimality of B_{T+k} codebooks, all T)**.** *If K is B_{T+k} and $d \ge T + k$, then $P_{\mathrm{tor}} \le P_K(T, k) \le W(d, T, k) \prod_{i=0}^{T+k-1} \big(1 + \frac{i}{d}\big)$.*

Proof. Under B_{T+k}, $\mathbb{E}_x \psi_x^{\otimes (T+k)}$ is diagonal in the occupation basis with entries $\mathrm{mult}(\beta)/d^{T+k}$, whose maximum is $(T+k)!/d^{T+k}$ for $d \ge T + k$. Apply Lemma 5 with $\dim V \le D_T$ and use $D_m = \prod_{i<m}(d+i)/m!$. □

4 The Sidon theorem

Theorem 7. *For every $K \subset \mathbb{Z}_q$ with $|K| = d \geq 2$, $P_K(1,1) = \dfrac{2d-1}{d^2}$ if and only if K is a Sidon set; otherwise $P_K(1,1) > (2d-1)/d^2$.*

Proof. Primal. The channel with Kraus operators $A_j|a\rangle = \sqrt{(2-\delta_{aj})/(2d-1)}\,|e_a+e_j\rangle_{\mathrm{Sym}^2}$, $j \in [d]$, is trace preserving and has fidelity $(2d-1)/d^2$ on every phase state, so $P_K(1,1) \geq (2d-1)/d^2$ for all K.

Dual. In Lemma 3 with $T = k = 1$, take $z_a \equiv d^3/(2d-1)$ (i.e. $Y = \frac{2d-1}{d^3} I$, trace $(2d-1)/d^2$). Torus sectors are indexed by charges $c = \beta - \alpha$. Type-1 charges $c = e_j$ have weight $\sum_a d^{-3}(2-\delta_{aj}) = (2d-1)/d^3$ and are tight; type-2 charges $c = e_b + e_{b'} - e_a$ ($a \notin \{b, b'\}$) have value $(2-\delta_{bb'})/(2d-1)$. For a Sidon set, a type-1 sector merges with nothing (two type-1 charges merge only if $k_i \equiv k_j$; type 1 with type 2 only through a non-trivial relation $k_j + k_a \equiv k_b + k_{b'}$). In a merged class of type-2 charges, Sidon makes the representation $s + k_a \equiv k_b + k_{b'}$ unique for each a, so the class value is at most $\sum_a (2 - \delta_{b_a b'_a})/(2d-1)$, which can exceed 1 only if all d edges $E_a = \{b_a, b'_a\}$ avoid a. If two such edges share a vertex, $E_a = \{u, v\}$, $E_{a'} = \{u, w\}$, subtracting gives $k_a + k_w \equiv k_{a'} + k_v$, and Sidon forces $a = v \in E_a$, a contradiction; hence the d edges are pairwise disjoint on d vertices, which is impossible. So Y is feasible.

Only if. We use the Lagrangian lower bound for the convex program of Lemma 3. With $T = k = 1$ and distinct keys, $r_1(w) = 1$ for $w \in K$, so $P_K(1,1) = \min_{z>0} \sum_a 1/z_a$ subject to $g_s(z) := d^{-3} \sum_a r_2(s + k_a) z_a \leq 1$ for all s. For any $\lambda \geq 0$ supported on $\{s \in K\}$ and any feasible z,

$$\sum_a \frac{1}{z_a} \;\geq\; \sum_a \frac{1}{z_a} + \sum_s \lambda_s \big(g_s(z) - 1\big) \;\geq\; \min_{z>0} \Big[\sum_a \Big(\frac{1}{z_a} + c_a z_a \Big) \Big] - \sum_s \lambda_s = 2 \sum_a \sqrt{c_a} - \sum_s \lambda_s,$$

with $c_a = d^{-3} \sum_s \lambda_s r_2(s + k_a)$. Take $\lambda_s = \lambda := (2d-1)/d^3$ for every $s \in K$. For every set of distinct keys, $r_2(k_j + k_a) \geq 2 - \delta_{ja}$, so $c_a \geq \lambda(2d-1)/d^3 = \lambda^2$, with equality for all a exactly when K is Sidon (then the bound equals $2d\lambda - d\lambda = (2d-1)/d^2$). If K is not Sidon there are multisets $\{i, j\} \neq \{l, m\}$ with $k_i + k_j \equiv k_l + k_m$; then $r_2(k_i + k_j) > 2 - \delta_{ij}$, hence $c_i > \lambda^2$ and $P_K(1,1) \geq 2 \sum_a \sqrt{c_a} - d\lambda > (2d-1)/d^2$. □

Remark 8. The value $(2d-1)/d^2$ itself is the $1 \to 2$ torus optimum, known from [8] (and the $N = 1$ case of [9, Eq. (16)]); the content of Theorem 7 is the exact characterisation of the finite codebooks that attain it. For $d = 2$ it recovers the observation of [13] that three equally spaced equatorial qubit states already determine the phase-covariant cloner (there for single-copy fidelity). Our necessity statements concern *global* fidelity only.

5 The torus theorem

Theorem 9 (Torus theorem)**.** *Let K be a B_{T+k} set mod q. Then $P_K(T,k) = P_{\mathrm{tor}}(d;T,k)$ in each of the following cases: (a) $d = 2$, all $T, k \geq 1$; (b) $T \in \{1,2,3\}$, all $d \geq 2$, $k \geq 1$.*

5.1 Reduction to a packing inequality

Let C be the set of balanced vectors with entries $\lfloor k/d \rfloor$ or $\lfloor k/d \rfloor + 1$ summing to k, and for $|b| = T$ put $Z_b = \sum_{g \in C} \mathrm{mult}(b+g)$ and $a_b = \sqrt{\mathrm{mult}(b)/Z_b}$. For a shift $\Delta \in \mathbb{Z}^d$ with $\sum_i \Delta_i = k$ define

$$\Lambda(\Delta) = \sum_{|b|=T,\ b \geq \Delta^-} a_b \,\mathrm{mult}(b + \Delta), \qquad \Lambda_0 = \Lambda(g)\ (g \in C), \qquad \rho(F) = \frac{1}{\Lambda_0} \sum_{\Delta \in F} \Lambda(\Delta).$$

Lemma 10 (Separation)**.** *Let K be a B_h set mod q and let $w \in \mathbb{Z}^d$, $w \neq 0$, satisfy $\sum_i w_i = 0$ and $\sum_i k_i w_i \equiv 0 \pmod q$. Then $\|w\|_1 \geq 2(h+1)$.*

Proof. Since $\sum_i w_i = 0$, the positive and negative parts $w^{\pm} \geq 0$ satisfy $|w^+| = |w^-| = \|w\|_1/2$, and they have disjoint supports, so they are distinct multisets on $[d]$ ($w \neq 0$). If $\|w\|_1 \leq 2h$, add the same multiset of size $h - |w^+|$ (e.g. copies of index 1) to both; this gives two distinct multisets of size h whose K-sums differ by $\sum_i k_i w_i \equiv 0$, contradicting B_h. □

Proposition 11. *If K is B_{T+k}, two distinct torus charges merged into the same class of Lemma 3 satisfy $\|\Delta - \Delta'\|_1 \geq 2(T+k+1)$. Consequently, if every family F of shifts $\Delta \in \mathbb{Z}^d$ with $\sum_i \Delta_i = k$ and $|\Delta^-| \leq T$ (the charges $\beta - \alpha$ with $|\alpha| = T$, $|\beta| = T+k$), pairwise $\|\Delta - \Delta'\|_1 \geq 2(T+k+1)$ and $|F| \geq 2$ satisfies $\rho(F) \leq 1$, then the torus dual point $z^\star_\alpha = d^{2T+k} a_\alpha / \Lambda_0$ (which corresponds to the dual point of [9, Eq. (147)] via $y_\alpha = \mathrm{mult}(\alpha)/z_\alpha$) is feasible for K, and $P_K(T,k) = P_{\mathrm{tor}}$.*

Proof. Classes. B_{T+k} implies B_T, so each class $\{\alpha : K \cdot \alpha \equiv w\}$ of Lemma 3 is a single multiset α, with $r_T(w) = \mathrm{mult}(\alpha)$; likewise $r_{T+k}(v) = \mathrm{mult}(\beta)$ for the unique β with $K \cdot \beta \equiv v$, if any. The constraint of class s therefore reads $\sum_{(\alpha,\beta): K\cdot(\beta-\alpha) \equiv s} \mathrm{mult}(\beta)\, z_\alpha \leq d^{2T+k}$. Group the pairs by their charge $\Delta = \beta - \alpha$; the charges occurring form the family $F_s = \{\Delta : K \cdot \Delta \equiv s\}$, and every α with $\alpha + \Delta \geq 0$ occurs with each $\Delta \in F_s$. Two distinct $\Delta, \Delta' \in F_s$ have $w = \Delta - \Delta' \neq 0$, $\sum w_i = 0$ and $K \cdot w \equiv 0$, so $\|\Delta - \Delta'\|_1 \geq 2(T+k+1)$ by Lemma 10. The class-s constraint is $\sum_{\Delta \in F_s} \sum_{\alpha \geq \Delta^-} \mathrm{mult}(\alpha + \Delta)\, z_\alpha \leq d^{2T+k}$.

Torus dual. Put $z^\star_\alpha = d^{2T+k} a_\alpha / \Lambda_0$. Then the class-$s$ constraint at $z^\star$ is exactly $\rho(F_s) \leq 1$. For $|F_s| = 1$ this is $\Lambda(\Delta) \leq \Lambda_0$, which is the feasibility of $z^\star$ in the torus problem (where every class is a single charge); for $|F_s| \geq 2$ it is the hypothesis. The objective is $\sum_\alpha \mathrm{mult}(\alpha)/z^\star_\alpha = d^{-(2T+k)} \Lambda_0 \sum_\alpha \sqrt{\mathrm{mult}(\alpha) Z_\alpha}$. Since a is S_d-invariant and C is a single S_d-orbit, $\Lambda(g) = \Lambda_0$ for all $g \in C$, and averaging over C gives $\Lambda_0 = |C|^{-1} \sum_\alpha a_\alpha Z_\alpha = |C|^{-1} \sum_\alpha \sqrt{\mathrm{mult}(\alpha) Z_\alpha}$. Hence the objective equals $|C| \Lambda_0^2 / d^{2T+k}$, which is exactly the optimal fidelity $F^\star_{\mathrm{ph}}(d,T,k) = P_{\mathrm{tor}}$ of [9, Eq. (16)]. Therefore $P_K(T,k) \leq P_{\mathrm{tor}}$, and Lemma 4 gives equality. □

Remark 12 (Identification with the dual certificate of [9]). In the notation of [9, Sec. X] ($N = T$, $K = k$, $\mathcal{C}_{d,K} = C$, $\mathcal{D}_\Delta = \{b : b + \Delta \geq 0\}$ by their Eq. (111)), the phase-covariant dual of their Prop. X.2 has variables $y_b > 0$ and, by their Eq. (119), one scalar constraint per shift, $\sum_{b \in \mathcal{D}_\Delta} |v_\Delta(b)|^2 / y_b \leq 1$ with $|v_\Delta(b)|^2 = d^{-(2T+k)} \mathrm{mult}(b)\, \mathrm{mult}(b+\Delta)$ (their Eq. (115)). Their quantities Z_b (Eq. (101)) and $\Lambda(\Delta)$ (Eq. (141)) coincide with ours. Under $y_b = \mathrm{mult}(b)/z_b$ this constraint is exactly the single-charge constraint $d^{-(2T+k)} \sum_{b \geq \Delta^-} \mathrm{mult}(b+\Delta) z_b \leq 1$ of Lemma 3, and the objectives agree, $\sum_b y_b = \sum_b \mathrm{mult}(b)/z_b$. Their dual point, Eq. (147), $y_b = \Lambda(\Gamma_0) \sqrt{\mathrm{mult}(b) Z_b} / d^{2T+k}$, corresponds to $z_b = \mathrm{mult}(b)/y_b = d^{2T+k} a_b / \Lambda_0 = z^\star_b$. Their Eq. (148) shows that the constraint of shift Δ at this point equals $\Lambda(\Delta)/\Lambda(\Gamma_0)$, and their Prop. X.5 (Eq. (146)) gives $\Lambda(\Delta) \leq \Lambda(\Gamma_0) = \Lambda_0$ for every integer shift with $|\Delta| = k$. Hence $z^\star$ is feasible for every single-charge class, which is the step used in the proof above; its objective is their Eq. (150), $|C| \Lambda_0^2 / d^{2T+k} = F^\star_{\mathrm{ph}}(d,T,k)$.

5.2 The packing inequality

Theorem 13. *Every family as in Proposition 11 with $|F| \geq 2$ satisfies: (a) $\rho(F) \leq 1$ for $d = 2$, all T, k; (b) $\rho(F) \leq \frac{d}{(k+1)(d-k/2)}$ for $T = 1$, $k \leq d-1$, and $\rho(F) \leq \frac{3}{4(d-1)}$ for $T = 1$, $k \geq d$; (c) $\rho(F) \leq 0.9961$ for $T = 2$, $d \geq 3$; (d) $\rho(F) \leq 0.9706$ for $T = 3$, $d \geq 3$.*

Parts (a) and (b) have complete pen-and-paper proofs (Appendix A). Parts (c) and (d) are computer-assisted: analytic bounds (normalisation Lemma N, structure Lemma S, multinomial

Table 1: Torus values $P_{\mathrm{tor}}(4;T,k)$, computed from [9, Eq. (16)] and independently from Lemma 3 (agreement $< 10^{-9}$), and Werner's universal value $W(4,T,k) = D_T/D_{T+k}$.

(T,k)	$P_{\mathrm{tor}}(4;T,k)$	$W(4,T,k)$
(1,1)	0.437500	0.400000
(1,2)	0.281250	0.200000
(2,1)	0.577816	0.500000
(1,3)	0.234375	0.114286
(2,2)	0.428776	0.285714
(3,1)	0.682631	0.571429

moves Lemma M, and a Johnson-type multiset packing bound Lemma J) reduce them to finitely many closed-form evaluations, which are verified in interval arithmetic, and to finitely many residual cases closed separately (13 at $T = 2$, by Lagrangian bounds evaluated in interval arithmetic; 18 at $T = 3$, by exact clique-LP certificates); see Appendix C. The decimal constants are outward-rounded upper bounds; every certificate is compared with 1 in exact rational or interval arithmetic.

Conjecture 14. *For all $d \geq 3$, $T \geq 4$ and $k \geq 1$, every family as in Proposition 11 with $|F| \geq 2$ satisfies $\rho(F) < 1$; in particular, $P_K(T,k) = P_{\mathrm{tor}}$ for every B_{T+k} set K.*

The conjecture is certified by exact clique-LP certificates for $T = 4$ ($3 \leq d \leq 10$, $k \leq 12$; $11 \leq d \leq 14$, $k \leq 10$), $T = 5$ ($3 \leq d \leq 9$, $k \leq 6$), all $d \leq 16$ with $T + k \leq 5$, and $d = 3$ with $T \leq 60$ ($k = 1$) and $T \leq 40$ ($k \leq 3$). On $d = 3$ the margin $1 - \max \rho$ decays like $\approx 2.9/T^2$, so a proof for all T must be sharp in T.

5.3 Negative results

Sidon alone does not suffice beyond $T + k = 2$. For $K = \{18, 21, 27, 28\} \subset \mathbb{Z}_{32}$ (Sidon, not B_3), $P_K(1,1) = 0.4375 = P_{\mathrm{tor}}$ but $P_K(1,2) = 0.291797 > P_{\mathrm{tor}} = 0.281250$ and $P_K(2,1) = 0.607913 > P_{\mathrm{tor}} = 0.577816$.

Table 2 extends this to $T + k \in \{3, 4\}$ and $d \in \{4, 6, 8\}$. In all 15 rows every set that is Sidon but not B_{T+k} (the Sidon-only sets and the Singer sets) has $P_K(T,k) > P_{\mathrm{tor}}$, and at fixed $T + k$ the margin grows with T. Singer sets, which are optimal for bias and form equiangular tight frames, are always worse than the Sidon-only sets and come within 0.017 of 1 at $(3,1)$, where they are the worst set tested for $d = 4, 6$. Arithmetic progressions, which have large bias, are the worst set in 11 of the 15 rows. Small Fourier bias therefore does not imply counterfeit resistance; what matters is the absence of additive coincidences of order $T + k$. These separations are numerical: the entries are reduced-dual values (for $d = 4$ also bracketed by a primal SDP), not certificates. The failure of Sidon-only sets is also not universal in d: for $d = 2$ and $(T,k) = (1,2)$, every Sidon set that is not B_3 among the 24 we tested (e.g. $K = \{0, 26\} \subset \mathbb{Z}_{39}$) attains $P_{\mathrm{tor}}(2;1,2) = 0.75$, whereas for $d = 2$, $(T,k) = (2,1)$ and for $d = 3$ at $T + k \in \{3, 4\}$ none of the tested sets does (24 and 72 sets, $q < 40$).

5.4 The case $T = 1$: an exact characterisation

Fix $k \geq 1$. A *charge* is a vector $c \in \mathbb{Z}^d$ with $\sum_i c_i = k$ and $c + e_a \geq 0$ for some $a \in [d]$; let $\mathcal{C}_k$ be the (finite) set of charges, and put

$$M(c) = \sum_{a \in [d]} \mathrm{mult}(c + e_a) \qquad (\mathrm{mult}(\gamma) := 0 \text{ if } \gamma \not\geq 0), \qquad M_k^\star = \max_{c \in \mathcal{C}_k} M(c).$$

Table 2: Counterfeiting values $P_K(T,k)$ at $T+k \in \{3,4\}$ for the sets of Table 3, from the reduced dual of Lemma 3 (QH-Lab `qhlab.lemma_r`). Bold: largest value in the row. A matched B_{T+k} set of the same size gives P_{tor} in every row. All entries agree with the independent R5 sweep to $\le 5\times 10^{-8}$; for $d=4$ they also agree with a primal (lower-bound) SDP to $\le 10^{-6}$.

d	(T,k)	$P_{\mathrm{tor}}(d;T,k)$	Sidon only	Singer	AP	random
4	(1,2)	0.281250	0.295129	0.343750	**0.686077**	0.434341
4	(2,1)	0.577816	0.624386	0.814044	**0.918002**	0.804521
4	(1,3)	0.234375	0.248486	0.331215	**0.604402**	0.409700
4	(2,2)	0.428776	0.519855	0.764730	0.875034	**0.907654**
4	(3,1)	0.682631	0.800184	**0.985848**	0.969445	0.937383
6	(1,2)	0.138889	0.168386	0.212963	**0.674107**	0.378659
6	(2,1)	0.421234	0.456406	0.715224	**0.930640**	0.776000
6	(1,3)	0.083333	0.141279	0.212560	**0.600597**	0.245909
6	(2,2)	0.232883	0.304882	0.674970	**0.881735**	0.669629
6	(3,1)	0.517533	0.714506	**0.983498**	0.976595	0.909976
8	(1,2)	0.082031	0.119188	0.152344	**0.669905**	0.368231
8	(2,1)	0.330058	0.445114	0.666175	**0.936120**	0.851925
8	(1,3)	0.038086	0.095927	0.148619	**0.599266**	0.309036
8	(2,2)	0.144007	0.363933	0.628594	**0.885440**	0.800783
8	(3,1)	0.413747	0.792293	0.984480	0.979150	**0.988168**

A charge with $M(c)=M_k^\star$ is *tight*. Charges $c\neq c'$ are *merged* by K if $K\cdot c\equiv K\cdot c' \pmod q$, where $K\cdot c=\sum_i c_i k_i$. Every charge has $M(c)\ge 1$. For $s\in\mathbb{Z}_q$ let $\Lambda_K(s)=\sum_{a\in[d]} r_{k+1}(s+k_a)$.

Lemma 15 (Aggregation). *$\Lambda_K(s)=\sum_{c\in\mathcal{C}_k:\,K\cdot c\equiv s} M(c)$ for every $s\in\mathbb{Z}_q$.*

Proof. $r_{k+1}(v)=\sum_{|\beta|=k+1,\,K\cdot\beta\equiv v}\mathrm{mult}(\beta)$. Substituting $c=\beta-e_a$, $\sum_a\sum_{K\cdot\beta\equiv s+k_a}\mathrm{mult}(\beta)=\sum_a\sum_{c:\,c+e_a\ge 0,\,K\cdot c\equiv s}\mathrm{mult}(c+e_a)=\sum_{c\in\mathcal{C}_k,\,K\cdot c\equiv s}M(c)$. □

Lemma 16 (Tight charges). *For all $d\ge 2$, $k\ge 1$, the tight charges in $\mathcal{C}_k$ are exactly the balanced vectors $c\ge 0$ with $|c|=k$ and entries in $\{\lfloor k/d\rfloor,\lfloor k/d\rfloor+1\}$; every other charge has $M(c)<M_k^\star$. In particular $M_k^\star=\mathrm{mult}(c^*)\sum_a\frac{k+1}{c_a^*+1}$ for any balanced c^*.*

Proof. Charges with a negative entry. Such a charge is $c=\beta-e_a$ with $|\beta|=k+1$ and $\beta_a=0$, so $M(c)=\mathrm{mult}(\beta)$. Pick b with $\beta_b\ge 1$. Then $c'=\beta-e_b\ge 0$ is a charge, and $M(c')\ge \mathrm{mult}(c'+e_b)+\mathrm{mult}(c'+e_a)=\mathrm{mult}(\beta)+\mathrm{mult}(\beta-e_b+e_a)>\mathrm{mult}(\beta)=M(c)$.

Nonnegative charges. For $c\ge 0$ with $|c|=k$ we have $M(c)=\mathrm{mult}(c)\sum_a\frac{k+1}{c_a+1}$. Suppose $c_i=x\ge y+2$ with $y=c_j$, and put $c'=c-e_i+e_j$ and $R=\sum_{a\neq i,j}\frac{1}{c_a+1}\ge 0$. Then $\mathrm{mult}(c')=\mathrm{mult}(c)\,\frac{x}{y+1}$, and

$$\frac{M(c')}{(k+1)\,\mathrm{mult}(c)}=\frac{1}{y+1}+\frac{x}{(y+1)(y+2)}+\frac{x}{y+1}R\;>\;\frac{1}{y+1}+\frac{1}{x+1}+R=\frac{M(c)}{(k+1)\,\mathrm{mult}(c)}.$$

This holds because $x/(y+1)>1$ and $x(x+1)>(y+1)(y+2)$. So any non-balanced $c\ge 0$ is strictly beaten by a less unbalanced charge. The balanced charges are permutations of one another, so they all have the same value. □

Proposition 17. *$P_{\mathrm{tor}}(d;1,k)=M_k^\star/d^{k+1}$. Moreover the program*

$$\min_{z\in(0,\infty)^d}\sum_a\frac{1}{z_a}\quad s.t.\quad d^{-(k+2)}\sum_a\mathrm{mult}(c+e_a)\,z_a\le 1\quad\forall c\in\mathcal{C}_k \tag{2}$$

Table 3: Key sets used in Table 2 (one set per d and type, the same for every (T,k) except Sidon-only and random, which were drawn per row in the R5 sweep). "Sidon only" sets are B_2 but not B_3; Singer sets are perfect difference sets, hence Sidon and not B_3; AP is $\{0,\dots,d-1\}$ with $q=2^{20}$ (no wrap-around).

d	(T,k)	type	$K \subset \mathbb{Z}_q$
4	(1,2)	Sidon only	$\{13,20,21,31\} \subset \mathbb{Z}_{32}$
4	all	Singer	$\{0,1,3,9\} \subset \mathbb{Z}_{13}$
4	all	AP	$\{0,1,2,3\} \subset \mathbb{Z}_{1048576}$
4	(1,2)	random	$\{1,7,8,11\} \subset \mathbb{Z}_{16}$
4	(2,1)	Sidon only	$\{5,10,27,29\} \subset \mathbb{Z}_{32}$
4	(2,1)	random	$\{6,8,9,11\} \subset \mathbb{Z}_{16}$
4	(1,3)	Sidon only	$\{0,4,12,27\} \subset \mathbb{Z}_{32}$
4	(1,3)	random	$\{6,8,9,11\} \subset \mathbb{Z}_{16}$
4	(2,2)	Sidon only	$\{7,8,10,19\} \subset \mathbb{Z}_{32}$
4	(2,2)	random	$\{0,10,12,14\} \subset \mathbb{Z}_{16}$
4	(3,1)	Sidon only	$\{3,10,22,25\} \subset \mathbb{Z}_{32}$
4	(3,1)	random (Sidon)	$\{0,7,10,12\} \subset \mathbb{Z}_{16}$
6	(1,2)	Sidon only	$\{1,29,50,110,116,117\} \subset \mathbb{Z}_{128}$
6	all	Singer	$\{0,1,3,10,14,26\} \subset \mathbb{Z}_{31}$
6	all	AP	$\{0,1,2,3,4,5\} \subset \mathbb{Z}_{1048576}$
6	(1,2)	random	$\{0,11,15,17,21,29\} \subset \mathbb{Z}_{32}$
6	(2,1)	Sidon only	$\{4,45,67,72,95,125\} \subset \mathbb{Z}_{128}$
6	(2,1)	random	$\{1,5,6,17,29,30\} \subset \mathbb{Z}_{32}$
6	(1,3)	Sidon only	$\{11,12,99,117,123,125\} \subset \mathbb{Z}_{128}$
6	(1,3)	random	$\{5,9,11,27,28,31\} \subset \mathbb{Z}_{32}$
6	(2,2)	Sidon only	$\{14,25,30,60,82,121\} \subset \mathbb{Z}_{128}$
6	(2,2)	random	$\{2,4,5,23,25,29\} \subset \mathbb{Z}_{32}$
6	(3,1)	Sidon only	$\{0,10,23,43,91,126\} \subset \mathbb{Z}_{128}$
6	(3,1)	random	$\{5,9,13,15,22,31\} \subset \mathbb{Z}_{32}$
8	(1,2)	Sidon only	$\{51,53,59,83,93,104,121,122\} \subset \mathbb{Z}_{128}$
8	all	Singer	$\{0,1,3,13,32,36,43,52\} \subset \mathbb{Z}_{57}$
8	all	AP	$\{0,1,2,3,4,5,6,7\} \subset \mathbb{Z}_{1048576}$
8	(1,2)	random	$\{0,6,8,12,16,21,24,27\} \subset \mathbb{Z}_{32}$
8	(2,1)	Sidon only	$\{0,15,43,60,78,92,97,104\} \subset \mathbb{Z}_{128}$
8	(2,1)	random	$\{1,2,9,10,14,16,17,29\} \subset \mathbb{Z}_{32}$
8	(1,3)	Sidon only	$\{14,30,80,84,85,95,98,117\} \subset \mathbb{Z}_{128}$
8	(1,3)	random	$\{1,5,7,10,16,17,23,30\} \subset \mathbb{Z}_{32}$
8	(2,2)	Sidon only	$\{16,45,51,52,82,91,106,110\} \subset \mathbb{Z}_{128}$
8	(2,2)	random	$\{0,2,6,16,21,27,28,31\} \subset \mathbb{Z}_{32}$
8	(3,1)	Sidon only	$\{0,24,43,50,60,108,117,122\} \subset \mathbb{Z}_{128}$
8	(3,1)	random	$\{1,6,8,13,17,21,27,31\} \subset \mathbb{Z}_{32}$

has value $P_{\mathrm{tor}}(d;1,k)$ *and the unique minimiser* $z^\circ = (d^{k+2}/M_k^\star)\mathbf{1}$.

Proof. The feasible set of (2) is convex and invariant under permutations of $[d]$ (they permute $\mathcal{C}_k$), and the objective is strictly convex and symmetric. Hence the minimiser is unique (the midpoint of two minimisers would be strictly better) and permutation invariant, $z = \zeta\mathbf{1}$; such z is feasible iff $\zeta M_k^\star \le d^{k+2}$, so the value is $M_k^\star/d^{k+1}$. Now let $K' = \{1, b, b^2, \dots, b^{d-1}\}$ with $b = k+3$, and $q' > (k+2)b^{d-1}$. Sums of $(k+2)$-multisets from K' have base-b digits $\le k+2$, so K' is B_{k+2} mod q', and two charges $c = \beta - e_a$, $c' = \beta' - e_{a'}$ with $K'\cdot c \equiv K'\cdot c'$ satisfy $K'\cdot(\beta + e_{a'}) \equiv K'\cdot(\beta' + e_a)$, hence $\beta + e_{a'} = \beta' + e_a$ and $c = c'$. By Lemmas 3 and 15, $P_{K'}(1,k)$ is therefore exactly (2). Since K' is B_{k+1}, Theorem 9(b) gives $P_{K'}(1,k) = P_{\mathrm{tor}}(d;1,k)$. □

Theorem 18 ($T = 1$ characterisation)**.** *For every* $K \subset \mathbb{Z}_q$ *with* $|K| = d \ge 2$ *and every* $k \ge 1$,

$$P_K(1,k) = P_{\mathrm{tor}}(d;1,k) \iff \max_{s\in\mathbb{Z}_q} \Lambda_K(s) \le M_k^\star,$$

and otherwise $P_K(1,k) > P_{\mathrm{tor}}(d;1,k)$. *Equivalently, equality holds iff no class* $\{c \in \mathcal{C}_k : K\cdot c \equiv s\}$ *has total weight* $\sum M(c)$ *above* $M_k^\star$.

Proof. With $T = 1$ and distinct keys, $r_1 = \mathbf{1}_K$, so by Lemmas 3 and 15, $P_K(1,k)$ is the minimum of $\sum_a 1/z_a$ subject to $d^{-(k+2)}\sum_{c:\,K\cdot c\equiv s}\sum_a \mathrm{mult}(c+e_a)z_a \le 1$ for all s. Each such constraint is a sum of nonnegative terms, one per charge in the class, so every K-feasible z is feasible for (2); this gives $P_K(1,k) \ge P_{\mathrm{tor}}$ (Lemma 4). If equality holds, the K-optimal z, which exists by Lemma 3, minimises (2), so $z = z^\circ$ by Proposition 17. Then K-feasibility of z° reads $\Lambda_K(s) \le M_k^\star$ for all s. Conversely, if this holds, z° is K-feasible with value P_{tor}. □

Proposition 19 (Lagrangian form of necessity)**.** *Let* $\mathcal{T}$ *be the set of tight charges and, for* $\mu \ge 0$, $\lambda_s = \mu\,\#\{t \in \mathcal{T} : K\cdot t \equiv s\}$. *Then*

$$P_K(1,k) \;\ge\; 2\sum_a \sqrt{c_a} - \mu|\mathcal{T}|, \qquad c_a = d^{-(k+2)}\sum_s \lambda_s r_{k+1}(s+k_a) \;\ge\; c^\circ := \frac{\mu|\mathcal{T}|M_k^\star}{d^{k+3}},$$

with equality $c_a = c^\circ$ *for all* a *iff no tight charge is merged with another charge. At* $\mu|\mathcal{T}| = M_k^\star/d^{k+1}$ *the right-hand side equals* $P_{\mathrm{tor}}(d;1,k)$ *when no tight charge is merged, and exceeds it otherwise.*

Proof. The first inequality is the Lagrangian bound in the proof of Theorem 7, with r_2 replaced by r_{k+1}. By Lemma 15 applied to the single index a, $r_{k+1}(K\cdot t + k_a) = \sum_{c:\,K\cdot c\equiv K\cdot t}\mathrm{mult}(c+e_a) \ge \mathrm{mult}(t+e_a)$, with strict inequality iff the class of t contains some $c \ne t$ with $c + e_a \ge 0$. So $c_a \ge d^{-(k+2)}\mu\sum_{t\in\mathcal{T}}\mathrm{mult}(t+e_a)$. The last sum does not depend on a, because $\mathcal{T}$ is permutation invariant, and summing over a gives $|\mathcal{T}|M_k^\star$, so it equals $|\mathcal{T}|M_k^\star/d$. If a tight t is merged with some $c \ne t$, pick a with $c + e_a \ge 0$; then $c_a > c^\circ$. Finally, with $u = \mu|\mathcal{T}|$, $2d\sqrt{c^\circ} - u$ is maximised at $u = M_k^\star/d^{k+1}$, where it equals $M_k^\star/d^{k+1} = P_{\mathrm{tor}}$. □

Corollary 20 (Necessity of the merging condition)**.** *If some tight charge is merged with another charge, then* $P_K(1,k) > P_{\mathrm{tor}}(d;1,k)$.

Proof. The class of that tight charge has weight at least $M_k^\star + 1$; apply Theorem 18, or Proposition 19 for an explicit lower bound. □

Say that K satisfies condition (H_k) if no tight charge is merged with another charge. By Corollary 20, (H_k) is necessary for $P_K(1,k) = P_{\mathrm{tor}}$. It is also sufficient in the following cases.

Lemma 21. *Let* $N \ge 2$, $m = \lfloor N/2 \rfloor$, *and* $u, v \ge 0$ *with* $u + v \le m - 1$. *Then* $\binom{N}{u} + \binom{N}{v} \le \binom{N}{m}$.

Proof. Assume $u \le v$ and put $t = m-1-u$, so $u \le v \le t$ and $\binom{N}{v} \le \binom{N}{t}$. Further, $\binom{N}{t+1} - \binom{N}{t} = \binom{N}{t}\frac{N-2t-1}{t+1} \ge \binom{N}{t}\frac{2u+1}{m-u}$, since $N-2t-1 \ge 2u+1$ and $t+1 = m-u$. If $t \ge u+1$, then $\binom{N}{t} \ge \binom{N}{u+1} = \binom{N}{u}\frac{N-u}{u+1}$, and $\frac{(2u+1)(N-u)}{(u+1)(m-u)} \ge 1$. If $t = u$, then $m = 2u+1$ and $\frac{2u+1}{m-u} \ge 1$. In both cases $\binom{N}{t+1} - \binom{N}{t} \ge \binom{N}{u}$, so $\binom{N}{u} + \binom{N}{v} \le \binom{N}{t+1} \le \binom{N}{m}$, because $t+1 \le m$. □

Theorem 22 (Sufficiency of (H_k) in three regimes)**.** *$P_K(1,k) = P_{\mathrm{tor}}(d;1,k)$ iff (H_k) holds, in each of the following cases.*

(a) $k = 1$, all d: here (H_1) is the Sidon property (Theorem 7).

(b) $d = 2$, all k: here (H_k) holds iff $n := \mathrm{ord}(k_2 - k_1) \ge \lceil k/2 \rceil + 2$.

(c) $k = 2$, all d: here (H_2) holds iff the only coincidences among sums of 3-multisets from K are $3k_i \equiv 3k_l$, i.e. every 3-multiset with two distinct elements has a unique sum.

Proof. Necessity is Corollary 20; we prove sufficiency through Theorem 18.

(a) For $k = 1$, $M(e_j) = 2d-1$, and every other charge $c = e_b + e_{b'} - e_a$ has $M(c) \le 2 < 2d-1$. If $k_i + k_j \equiv k_l + k_m$ with $\{i,j\} \ne \{l,m\}$, then the tight charge e_i is merged with $e_l + e_m - e_j \ne e_i$, so (H_1) implies Sidon. For Sidon K, the dual part of the proof of Theorem 7 shows that $\Lambda_K \le 2d-1$.

(b) Let $N = k+2$ and $g = k_2 - k_1$. The charges are $c = (k-j, j)$, $j = -1, \dots, k+1$, with $M(c) = \binom{k+1}{j} + \binom{k+1}{j+1} = \binom{N}{i}$, where $i = j+1 \in [0, N]$, and $K \cdot c = kk_1 + jg$. So two charges are merged iff their indices i agree mod n. The tight indices are $\lfloor N/2 \rfloor$ and $\lceil N/2 \rceil$, so (H_k) holds iff $n > \lceil N/2 \rceil$, i.e. $n \ge \lceil k/2 \rceil + 2$. Then every class has at most two indices $i < i+n$, and with $v = N - i - n$ we get $i + v = N - n \le \lfloor N/2 \rfloor - 1$. Lemma 21 gives $\binom{N}{i} + \binom{N}{i+n} = \binom{N}{i} + \binom{N}{v} \le \binom{N}{\lfloor N/2 \rfloor} = M_k^\star$.

(c) Let $d \ge 3$ ($d = 2$ is covered by (b)). The weights are $M(e_i + e_j) = 6d - 6$ for $i \ne j$, $M(2e_i) = 3d-2$, and $M(\beta - e_a) = \mathrm{mult}(\beta) \in \{1, 3, 6\}$ for $|\beta| = 3$, $\beta_a = 0$. So the tight charges are exactly the $e_i + e_j$, $i \ne j$. A tight $e_i + e_j$ is merged with $c' = \beta - e_a$ ($\beta \ge 0$) iff $K \cdot (e_i + e_j + e_a) \equiv K \cdot \beta$ with $\beta \ne e_i + e_j + e_a$. Every 3-multiset with two distinct elements has the form $\{i, j, a\}$ with $i \ne j$, so this proves the stated form of (H_2).

Assume (H_2) and fix a class S; we show $\sum_{c \in S} M(c) \le 6d - 6$.

- If S contains a tight charge, S is that charge alone.
- If $S \ni 2e_i$, consider any other $c' \in S$. It is not tight. It is not $2e_l$: from $2k_i \equiv 2k_l$ we would get $\{i, i, x\} \sim \{l, l, x\}$, contradicting ($\mathrm{H}_2$). So $c' = \beta - e_a$ with $\beta_a = 0$. Then $2e_i + e_a$ and β are distinct 3-multisets with equal sums, so both are pure, which forces $a = i$ and $\beta = 3e_l$. Hence $\sum_S M \le 3d - 2 + (d-1) = 4d - 3 \le 6d - 6$.
- Otherwise every member has the form $c_\nu = \beta_\nu - e_{a_\nu}$ with $\beta_\nu(a_\nu) = 0$, and this representation is unique. We use three facts.
  (1) If $a_\nu = a_\mu$ for $\nu \ne \mu$, then $K \cdot \beta_\nu \equiv K \cdot \beta_\mu$, so both β's are pure.
  (2) If β_ν has three distinct elements, then for every $\mu \ne \nu$ the 4-multisets $\beta_\nu + e_{a_\mu}$ and $\beta_\mu + e_{a_\nu}$ are disjoint. They have equal sums and are distinct. A common element x could be removed, and this would leave distinct 3-multisets with equal sums; the first of these has at least two distinct elements, contradicting (H_2). So the supports of β_ν and β_μ are disjoint.
  (3) For each l there is at most one member $3e_l - e_a$, since a is determined by $k_a \equiv 3k_l - s$.

Let x, y, w be the numbers of members with $\mathrm{mult}(\beta) = 6, 3, 1$, and let U be the union of the supports of the type-6 β's; by (2) these supports are pairwise disjoint, so $|U| = 3x \le d$. By (1), two members with the same index a both have pure β's, so the type-6 and type-3 members have pairwise distinct indices a_ν, and $x + y \le d$. For a type-1 member μ, $\beta_\mu = 3e_l$ has support $\{l\}$, so (2), applied to each type-6 member ν and this μ, gives $l \notin \operatorname{supp} \beta_\nu$; hence $l \notin U$. By (3) these l are pairwise distinct, so $w \le d - 3x$. Hence $\sum_S M = 6x + 3y + w \le 6x + 3(d - x) + (d - 3x) = 4d \le 6d - 6$, since $d \ge 3$.

□

Conjecture 23. *For all $d \ge 2$ and $k \ge 1$: $P_K(1,k) = P_{\mathrm{tor}}(d;1,k)$ iff (H_k) holds.*

Remark 24 (The $d = 2$ exception and $d \ge 3$). Theorem 22 explains the $d = 2$, $(T,k) = (1,2)$ observation of Section 5.3. For $d = 2$, B_3 means $n \ge 4$, but $P_K(1,2) = P_{\mathrm{tor}}$ already holds iff $n \ge 3$. Thus $\{0, 26\} \subset \mathbb{Z}_{39}$ ($n = 3$) attains 0.75, and in general $d = 2$ sets with $\lceil k/2 \rceil + 2 \le n \le k + 1$ attain $P_{\mathrm{tor}}(2;1,k)$ without being B_{k+1}. The mechanism is not specific to $d = 2$: by (c), a Sidon set whose only 3-relations are $3k_i \equiv 3k_l$ attains $P_{\mathrm{tor}}(d;1,2)$. Examples are $\{0,1,13\} \subset \mathbb{Z}_{39}$ ($d = 3$, value 4/9), $\{0,10,11,14\} \subset \mathbb{Z}_{30}$ ($d = 4$, 9/32) and $\{0,23,34,36,50\} \subset \mathbb{Z}_{69}$ ($d = 5$, 24/125), none of them B_3. Similarly $\{0,10,26,29\} \subset \mathbb{Z}_{30}$ attains $P_{\mathrm{tor}}(4;1,3) = 15/64$ without being B_3. So B_{T+k} is *not* necessary for $P_K = P_{\mathrm{tor}}$ when $d \ge 3$, at least for $T = 1$. The Sidon-only sets in the tables of Section 5.3 fail because they carry a non-pure 3-relation; for example $\{18,21,27,28\} \subset \mathbb{Z}_{32}$ has $2 \cdot 18 + 27 \equiv 3 \cdot 21$, a tight charge is merged, and $\max_s \Lambda_K(s) = 19 > 18$.

Remark 25 (No finite codebook works for all k). Since $\sum_s \Lambda_K(s) = d^{k+2}$, we have $\max_s \Lambda_K(s) \ge d^{k+2}/q$. On the other hand, $M_k^\star/d^{k+2} \le \max_\beta \mathrm{mult}(\beta)/d^{k+1} \to 0$ as $k \to \infty$, since this is the largest point probability of a uniform multinomial. Hence for every finite $K \subset \mathbb{Z}_q$, $P_K(1,k) > P_{\mathrm{tor}}(d;1,k)$ for all sufficiently large k; precisely, this holds as soon as $P_{\mathrm{tor}}(d;1,k) < d/q$.

6 Fusion and additive energy

Theorem 26 (Fusion). *For x, y independent and uniform on $\mathbb{Z}_q$, the optimal average fidelity of a channel $\psi_x \otimes \psi_y \mapsto \psi_{x+y}$ is $F_{\mathrm{fuse}}(K) = E_q(K)/d^3$. In particular $F_{\mathrm{fuse}}(K) \ge (2d-1)/d^2$, with equality iff K is Sidon.*

Proof. Let $r(v) = \#\{(a,b) : k_b - k_a \equiv v\}$, so $\sum_v r(v)^2 = E_q(K)$, and for each v let $S_v = \{j : k_j + v \in K\}$, $|S_v| = r(v)$, with $l(j,v)$ the index of $k_j + v$.

Achievability. Define $A_v = \sum_{j \in S_v} |j\rangle\langle j, l(j,v)|$. Each basis pair $|j,l\rangle$ lies in exactly one branch ($v = k_l - k_j$), so $\sum_v A_v^\dagger A_v = I$ and $\{A_v\}$ is a channel. On the input,

$$A_v(\psi_x \otimes \psi_y) = \frac{1}{d} \sum_{j \in S_v} \omega^{x k_j + y(k_j + v)} |j\rangle = \frac{\omega^{yv}}{d} \sum_{j \in S_v} \omega^{(x+y)k_j} |j\rangle.$$

The y-dependent factor is a global phase of the branch, so $\left|\langle \psi_{x+y} | A_v(\psi_x \otimes \psi_y)\right|^2 = r(v)^2/d^3$, and the output fidelity is $\sum_v r(v)^2/d^3 = E_q(K)/d^3$ for *every* (x,y).

Optimality. The Choi operator of the target ensemble is $X = \mathbb{E}_{x,y}\, \bar{\psi}_x \otimes \bar{\psi}_y \otimes \psi_{x+y}$, with entries $X_{(jlm),(j'l'm')} = d^{-3}\,[k_m - k_j \equiv k_{m'} - k_{j'}]\,[k_m - k_l \equiv k_{m'} - k_{l'}]$. Hence $X = d^{-3} \sum_{(s,t)} |\chi_{st}\rangle\langle\chi_{st}|$, where χ_{st} is the 0/1 indicator of $B_{st} = \{(j,l,m) : k_m - k_j \equiv s,\ k_m - k_l \equiv t\}$; the blocks B_{st} partition the basis, and on B_{st} the difference $k_l - k_j \equiv s - t =: v$ is constant. Let $Y = \mathrm{diag}_{(j,l)}\, r(k_l - k_j)/d^3$. Since $Y \otimes I$ is diagonal and positive on each block, $Y \otimes I \succeq d^{-3} |\chi_{st}\rangle\langle\chi_{st}|$ iff $\sum_{B_{st}} d^{-3}/(r(v)/d^3) \le 1$, i.e.

$|B_{st}| \le r(v)$. The map $(j,l,m) \mapsto (l,j)$ sends B_{st} injectively into $\{(a,b) : k_b - k_a \equiv -v\}$, whose size is $r(-v) = r(v)$. So Y is dual feasible, and $\operatorname{tr} Y = \sum_{(j,l)} r(k_l - k_j)/d^3 = \sum_v r(v)^2/d^3 = E_q(K)/d^3$.

Finally $E_q(K) \ge 2d^2 - d$ (the trivial solutions $a = b, c = e$ and $a = c, b = e$), with equality iff K is Sidon, giving the corollary. □

For $m \ge 2$ let $r(v) = \#\{(a,b) \in [d]^2 : k_b - k_a \equiv v\}$ as in the proof of Theorem 26, and define the *higher energy*

$$E_m(K) = \sum_{v \in \mathbb{Z}_q} r(v)^m,$$

so that $E_2(K) = E_q(K)$. For $j = (j_1, \dots, j_m) \in [d]^m$ put $\delta(j) = (k_{j_2} - k_{j_1}, \dots, k_{j_m} - k_{j_1}) \in \mathbb{Z}_q^{m-1}$ and, for $\delta = (\delta_2, \dots, \delta_m) \in \mathbb{Z}_q^{m-1}$ (with $\delta_1 := 0$),

$$N(\delta) = \#\{j \in [d]^m : \delta(j) = \delta\} = \#\{a \in [d] : k_a + \delta_i \in K \text{ for all } i\},$$

the second form because the keys are distinct, so j is determined by j_1 and $\delta(j)$.

Lemma 27. $\sum_{\delta \in \mathbb{Z}_q^{m-1}} N(\delta)^2 = E_m(K)$.

Proof. $\sum_\delta N(\delta)^2$ counts triples (a,b,δ) with $k_a + \delta_i \in K$ and $k_b + \delta_i \in K$ for $i = 2, \dots, m$. For fixed (a,b) the coordinates δ_i are chosen independently from $(K - k_a) \cap (K - k_b)$, a set of size $\#\{c \in K : c - k_a + k_b \in K\} = r(k_b - k_a)$. Hence the count is $\sum_{a,b} r(k_b - k_a)^{m-1} = \sum_v r(v) \cdot r(v)^{m-1} = E_m(K)$. □

Theorem 28 (m-fold fusion)**.** *Let $m \ge 2$ and let $x_1, \dots, x_m$ be independent and uniform on $\mathbb{Z}_q$. The optimal average fidelity of a channel $(\mathbb{C}^d)^{\otimes m} \to \mathbb{C}^d$ for $\psi_{x_1} \otimes \cdots \otimes \psi_{x_m} \mapsto \psi_{x_1 + \cdots + x_m}$ is*

$$F_m(K) = \frac{E_m(K)}{d^{m+1}},$$

and it is attained by a channel whose fidelity equals $F_m(K)$ on every *input $(x_1, \dots, x_m)$. For $m = 2$ this is Theorem 26.*

Proof. Write $X_\Sigma = x_1 + \cdots + x_m$.

Achievability. For $\delta \in \mathbb{Z}_q^{m-1}$ let $A_\delta = \sum_{j :\, \delta(j) = \delta} |j_1\rangle\langle j|$. Every basis vector $|j\rangle$ of $(\mathbb{C}^d)^{\otimes m}$ lies in exactly one branch, and distinct j in one branch have distinct j_1, so $\sum_\delta A_\delta^\dagger A_\delta = I$ and $\{A_\delta\}$ is a channel. Since $k_{j_i} = k_{j_1} + \delta_i$ on the branch,

$$A_\delta\big(\psi_{x_1} \otimes \cdots \otimes \psi_{x_m}\big) = d^{-m/2} \sum_{j :\, \delta(j) = \delta} \omega^{\sum_i x_i k_{j_i}} |j_1\rangle = d^{-m/2}\, \omega^{\sum_i x_i \delta_i} \sum_{j :\, \delta(j) = \delta} \omega^{X_\Sigma k_{j_1}} |j_1\rangle.$$

The factor $\omega^{\sum_i x_i \delta_i}$ is a global phase of the branch, so $\big|\langle \psi_{X_\Sigma} | A_\delta(\psi_{x_1} \otimes \cdots \otimes \psi_{x_m})\big|^2 = N(\delta)^2/d^{m+1}$, and the fidelity is $\sum_\delta N(\delta)^2/d^{m+1} = E_m(K)/d^{m+1}$ for every input (Lemma 27).

Optimality. The average fidelity of a channel with Choi operator J ($\operatorname{tr}_{\text{out}} J = I$) is $\operatorname{tr}(JX)$ with $X = \mathbb{E}_x\, \bar\psi_{x_1} \otimes \cdots \otimes \bar\psi_{x_m} \otimes \psi_{X_\Sigma}$, and $\operatorname{tr}(JX) \le \operatorname{tr}\big(J(Y \otimes I)\big) = \operatorname{tr} Y$ whenever $Y \otimes I \succeq X$. Averaging $\omega^{x_i t}$ over x_i gives

$$X_{(j,l),(j',l')} = d^{-(m+1)} \prod_{i=1}^{m} \big[k_l - k_{j_i} \equiv k_{l'} - k_{j_i'}\big].$$

With $\sigma(j,l) = (k_l - k_{j_1}, \dots, k_l - k_{j_m}) \in \mathbb{Z}_q^m$ and $B_\sigma = \{(j,l) : \sigma(j,l) = \sigma\}$, the blocks B_σ partition the basis and $X = d^{-(m+1)} \sum_\sigma |\chi_\sigma\rangle\langle\chi_\sigma|$, where χ_σ is the 0/1 indicator of B_σ. On B_σ the input difference vector is constant, $\delta(j) = (\sigma_1 - \sigma_i)_{i=2}^m =: \delta_\sigma$, and

$$|B_\sigma| = \#\{l : k_l - \sigma_i \in K \ \forall i\} = \Big|\bigcap_i (K + \sigma_i)\Big| = \Big|\bigcap_i (K - (\delta_\sigma)_i)\Big| = N(\delta_\sigma),$$

after translating by $-\sigma_1$. Let $Y = \operatorname{diag}_j N(\delta(j))/d^{m+1}$, which is positive since $N(\delta(j)) \geq 1$. Because $Y \otimes I$ is diagonal and X is a direct sum over the blocks, $Y \otimes I \succeq X$ holds iff on every block $\sum_{(j,l)\in B_\sigma} d^{-(m+1)} / \big(N(\delta_\sigma)/d^{m+1}\big) \leq 1$, i.e. $|B_\sigma| \leq N(\delta_\sigma)$, which holds with equality. Hence Y is dual feasible and $\operatorname{tr} Y = \sum_j N(\delta(j))/d^{m+1} = \sum_\delta N(\delta)^2/d^{m+1} = E_m(K)/d^{m+1}$. □

Corollary 29. *For $m \geq 2$, $E_m(K) \geq d^m + d^2 - d$, with equality if and only if K is a Sidon set. Hence $F_m(K) \geq (d^m + d^2 - d)/d^{m+1}$, with equality iff K is Sidon. Moreover $F_m(K) \leq 1$, with equality iff K is a coset of a subgroup of $\mathbb{Z}_q$ of order d.*

Proof. $r(0) = d$ and $\sum_{v\neq 0} r(v) = d^2 - d$. For integers $r \geq 0$ and $m \geq 2$, $r^m \geq r$ with equality iff $r \in \{0, 1\}$, so $E_m(K) = d^m + \sum_{v\neq 0} r(v)^m \geq d^m + d^2 - d$, with equality iff $r(v) \leq 1$ for all $v \neq 0$. This is the Sidon property: if $k_a + k_e \equiv k_b + k_c$ with $\{a, e\} \neq \{b, c\}$, then the ordered pairs (b, a) and (e, c) are distinct and have the same nonzero difference $k_a - k_b \equiv k_c - k_e$; conversely two distinct ordered pairs $(a, b) \neq (c, e)$ with $k_b - k_a \equiv k_e - k_c \not\equiv 0$ give $k_b + k_c \equiv k_a + k_e$ with $\{b, c\} \neq \{a, e\}$. For the upper bound, $r(v) \leq d$ gives $E_m(K) \leq d^{m-1} \sum_v r(v) = d^{m+1}$, with equality iff $r(v) \in \{0, d\}$ for all v, i.e. $K + v = K$ for every $v \in K - K$, i.e. K is a coset of a subgroup of order d. □

Remark 30. For $m \geq 3$ the optimal measurement reads out the full difference vector $\delta \in \mathbb{Z}_q^{m-1}$, and the optimum is the *common-base-point* count $\sum_\delta N(\delta)^2$; Lemma 27 identifies it with $\sum_v r(v)^m$. For $m = 3$ and $K = \{0, 1, 3\} \subset \mathbb{Z}_7$ (Sidon) this gives 33/81, for $\{0, 1, 2\} \subset \mathbb{Z}_7$ it gives 45/81, and for $\{0, 1, 2, 4\} \subset \mathbb{Z}_{11}$ it gives 100/256. In the notation of [14, 15], $r(v) = (K \circ K)(v) = |K \cap (K - v)|$, $E_m(K)$ is the higher energy $\mathsf{E}_m(K)$, and Lemma 27 is the identity $\mathsf{E}_m(A) = \sum_{s\in G^{m-1}} |A_s|^2$, $A_s = A \cap (A - s_1) \cap \cdots \cap (A - s_{m-1})$; see also the survey [16].

Thus the same additive structure that makes a codebook hard to clone makes it hard to fuse: Sidon sets minimise both, and perfect fusion forces orthogonal (hence clonable) states. Phase-state tokens are also malleable (the public shift Z^y maps $\psi_x \mapsto \psi_{x+y}$), and verification oracles leak the key [17].

7 Finite orbits of non-abelian groups

Let H be a compact group acting unitarily on V, $G \leq H$ closed, and $X = G \cdot f \subset H \cdot f$ with uniform weights. Because $X \subset H \cdot f$ and Haar measure is invariant, the H-twirl of R_X is R_H, i.e. $\int_H h\, D\, h^{-1} dh = 0$; this identity is used in parts (b) and (e) below. Let $u_x = \bar{x}^{(T)} \otimes x^{(T+k)}$, $R_X = \mathbb{E}_X u_x u_x^\dagger$, $R_H = \int_H R_{hf}\, dh$, $D = R_X - R_H$, and $P_\bullet = \min\{\operatorname{tr} Y : Y \otimes I \succeq R_\bullet\}$.

A natural guess, suggested by Proposition 6 and by the torus case, is that matching the $(T+k)$-th moment, $\mathbb{E}_X x^{\otimes(T+k)} = \mathbb{E}_H x^{\otimes(T+k)}$, implies $P_X = P_H$. It holds for unitary designs (Werner) and for orbits of highest-weight (coherent) vectors [18, 19], where Lemma 5 is tight. It fails in general:

Theorem 31 (Counterexamples)**.** *(i) $H = SU(3)$ acting on $V = \operatorname{Sym}^2 \mathbb{C}^3$ ($\dim V = 6$), G the Hessian group (order 216 mod phases), $(T, k) = (1, 1)$: there is a fiducial f with $M_2(X) = M_2(H)$ exactly and $P_X - P_H \geq 8.0399 \times 10^{-4}$. (ii) $H = SU(2)$ acting on spin 1 (through $SO(3)$, i.e. the vector representation on $\mathbb{C}^3$), G the octahedral rotation group (order 24), $(T, k) = (1, 1)$: there is a fiducial with $M_2(X) = M_2(H)$ exactly and $P_X - P_H \geq 6.437 \times 10^{-4}$.*

Both are certified in ball arithmetic, with exact moment matching established by a Poincaré–Miranda (resp. intermediate value) argument; see Appendix C. Independently re-implemented checks (exact Schur–Weyl twirl for $SU(3)$, exact Euler product rule for $SO(3)$) with other fiducials give strictly positive gaps in every tuned case (4/4 and 4/4).

Theorem 32 (Window theorem). *Assume (C1) G and H have the same commutant on* $\mathrm{Sym}^T V$. *Let $\mathcal{Y}_H$ be the H-invariant optimal duals for R_H; for $Y \in \mathcal{Y}_H$ let $Z_Y = Y \otimes I - R_H \succeq 0$ and Π_Y the projector onto* $\ker Z_Y$. *(a) $P_X \geq P_H$, with equality iff some $Y \in \mathcal{Y}_H$ has $Z_Y - D \succeq 0$. (b) If every G-invariant operator on* $\ker Z_Y$ *is H-invariant, then $\Pi_Y D \Pi_Y = 0$. (c) If the hypothesis of (b) holds for every $Y \in \mathcal{Y}_H$ and $(1-\Pi_Y)D\Pi_Y \neq 0$ for every $Y \in \mathcal{Y}_H$, then $P_X > P_H$. (Proof idea: $\Pi_Y(Z_Y - D)\Pi_Y = 0$, and a positive semidefinite operator with a vanishing diagonal block has vanishing off-diagonal blocks; so no optimal H-dual is feasible for X, and (a) applies.) (d) If some Y has $\Pi_Y D \Pi_Y = 0$, $(1-\Pi_Y)D\Pi_Y = 0$ and $\|(1-\Pi_Y)D(1-\Pi_Y)\| \leq \lambda^+_{\min}(Z_Y)$, then $P_X = P_H$. (e) (Selection rule) $(1-\Pi_Y)D\Pi_Y$ can only receive contributions from H-isotypic components $\lambda \neq \mathbf{1}$ of the moment difference $M_{2T+k}(X) - M_{2T+k}(H)$ with $V_\lambda^G \neq 0$ and $V_\lambda \hookrightarrow \mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp)$.*

Actions. Fix an orthonormal basis of V, write $\bar{A}$ for the entrywise complex conjugate and A^{t} for the transpose in this basis, and put $n = 2T + k$. Each $h \in H$ acts on

(A1) V by h, and on $\bar{V}$ (the same vector space with the conjugate representation) by $\bar{h}$. Since h is unitary, $\bar{h} = (h^{-1})^{\mathrm{t}}$, so $\bar{V} \cong V^*$ as H-modules;

(A2) $V^{\otimes n}$ by $h^{\otimes n}$; on $\bar{V}^{\otimes T} \otimes V^{\otimes(T+k)}$ by $\bar{h}^{\otimes T} \otimes h^{\otimes(T+k)}$; on the invariant subspaces $\mathrm{Sym}^T(\bar{V})$, $\mathrm{Sym}^{T+k}(V)$ and $\mathcal{K} = \mathrm{Sym}^T(\bar{V}) \otimes \mathrm{Sym}^{T+k}(V)$ by restriction. On $\mathcal{K}$ we write $U_h = \bar{h}^{\otimes T} \otimes h^{\otimes(T+k)}|_{\mathcal{K}}$, and on the input space $\mathrm{Sym}^T(\bar{V})$ of Y we write $\bar{h}_T = \bar{h}^{\otimes T}|_{\mathrm{Sym}^T(\bar{V})}$;

(A3) operators by conjugation: $\mathrm{Ad}_h(A) = h^{\otimes n} A h^{\dagger \otimes n}$ on $\mathrm{End}(V^{\otimes n})$, $A \mapsto U_h A U_h^\dagger$ on $\mathrm{End}(\mathcal{K})$, $Y \mapsto \bar{h}_T Y \bar{h}_T^\dagger$ on $\mathrm{End}(\mathrm{Sym}^T \bar{V})$, and, for H-invariant subspaces $\mathcal{A}, \mathcal{B} \subset \mathcal{K}$, $S \mapsto U_h|_{\mathcal{B}}\, S\, U_h|_{\mathcal{A}}^\dagger$ on $\mathrm{Hom}(\mathcal{A}, \mathcal{B})$; as H-modules $\mathrm{Hom}(\mathcal{A}, \mathcal{B}) \cong \mathcal{B} \otimes \mathcal{A}^*$.

The same formulas with $g \in G$ define the G-actions. An operator is *G-invariant* (resp. *H-invariant*) if it is fixed by all $g \in G$ (resp. $h \in H$). With $u_x = \bar{x}^{\otimes T} \otimes x^{\otimes(T+k)} \in \mathcal{K}$ we have $U_h u_x = u_{hx}$ and $U_h(Y \otimes I)U_h^\dagger = (\bar{h}_T Y \bar{h}_T^\dagger) \otimes I$. R_X is G-invariant because X is a G-orbit with uniform weights, and R_H is H-invariant by invariance of Haar measure. For every $x = gf \in X$, right invariance of Haar measure gives $\int_H U_h u_x u_x^\dagger U_h^\dagger \, dh = \int_H u_{hgf} u_{hgf}^\dagger \, dh = R_H$; averaging over X,

$$\int_H U_h R_X U_h^\dagger \, dh = R_H, \qquad \text{hence} \qquad \int_H U_h D U_h^\dagger \, dh = 0. \tag{3}$$

The minima defining P_X and P_H are attained: a feasible Y satisfies $Y \succeq 0$, and $\{Y \succeq 0 : \mathrm{tr}\, Y \leq c\}$ is compact. For $Y \in \mathcal{Y}_H$, $Z_Y = Y \otimes I - R_H$ is H-invariant, hence $\ker Z_Y$ and $(\ker Z_Y)^\perp$ are H-invariant subspaces of $\mathcal{K}$, Π_Y commutes with every U_h, and $Z_Y = (1-\Pi_Y)Z_Y(1-\Pi_Y)$. Finally $Z_Y - D = Y \otimes I - R_X$, and D is Hermitian.

Lemma 33 (Zero diagonal block). *Let $M \succeq 0$ on a finite-dimensional Hilbert space and let Π be an orthogonal projection with $\Pi M \Pi = 0$. Then $M\Pi = 0$ and $\Pi M = 0$; in particular $(1-\Pi)M\Pi = 0$.*

Proof. For every vector v,

$$0 = \langle \Pi v, M \Pi v \rangle = \|M^{1/2} \Pi v\|^2.$$

So $M^{1/2}\Pi = 0$, hence $M\Pi = M^{1/2}(M^{1/2}\Pi) = 0$ and $\Pi M = (M\Pi)^\dagger = 0$. □

Proof of Theorem 32. *(a)* Let Y attain P_X and put $Y_G = \int_G \bar{g}_T Y \bar{g}_T^\dagger \, dg$ (normalised Haar measure on G; the uniform average if G is finite). Conjugating $Y \otimes I \succeq R_X$ by U_g and using G-invariance of R_X gives $(\bar{g}_T Y \bar{g}_T^\dagger) \otimes I \succeq R_X$ for every g, so Y_G is feasible for X, and $\operatorname{tr} Y_G = \operatorname{tr} Y = P_X$. Y_G commutes with every $\bar{g}_T$; taking entrywise conjugates, $\overline{Y_G}$ commutes with every $g^{\otimes T}|_{\mathrm{Sym}^T V}$, so by (C1) it commutes with every $h^{\otimes T}|_{\mathrm{Sym}^T V}$, i.e. Y_G commutes with every $\bar{h}_T$: Y_G is H-invariant. Conjugating $Y_G \otimes I \succeq R_X$ by U_h gives $Y_G \otimes I \succeq U_h R_X U_h^\dagger$ for every h; integrating and using (3), $Y_G \otimes I \succeq R_H$. Hence $P_H \le \operatorname{tr} Y_G = P_X$. If $P_X = P_H$, then Y_G is an H-invariant optimal dual for R_H, i.e. $Y_G \in \mathcal{Y}_H$, and $Z_{Y_G} - D = Y_G \otimes I - R_X \succeq 0$. Conversely, if $Y \in \mathcal{Y}_H$ and $Z_Y - D = Y \otimes I - R_X \succeq 0$, then Y is feasible for X and $P_X \le \operatorname{tr} Y = P_H$; with $P_X \ge P_H$ this gives equality.

(b) The compression $B := \Pi_Y D \Pi_Y|_{\ker Z_Y} \in \mathrm{End}(\ker Z_Y)$ is G-invariant, because $D = R_X - R_H$ is G-invariant and Π_Y commutes with every U_g. By hypothesis B is H-invariant, so it equals its H-twirl: $\Pi_Y D \Pi_Y = \int_H U_h \Pi_Y D \Pi_Y U_h^\dagger \, dh = \Pi_Y \big(\int_H U_h D U_h^\dagger dh \big) \Pi_Y = 0$ by (3).

(c) Suppose $P_X = P_H$. By (a) some $Y \in \mathcal{Y}_H$ has $M := Z_Y - D \succeq 0$. Since $Z_Y \Pi_Y = 0$ and the hypothesis of (b) holds for this Y, $\Pi_Y M \Pi_Y = -\Pi_Y D \Pi_Y = 0$. By Lemma 33, $0 = (1-\Pi_Y) M \Pi_Y = (1-\Pi_Y) Z_Y \Pi_Y - (1-\Pi_Y) D \Pi_Y = -(1-\Pi_Y) D \Pi_Y$, contradicting the hypothesis $(1-\Pi_Y) D \Pi_Y \neq 0$. Hence $P_X \neq P_H$, and by (a) $P_X > P_H$.

(d) (Here $Y \in \mathcal{Y}_H$, as implicit in the notation Z_Y, Π_Y.) With respect to $\mathcal{K} = \ker Z_Y \oplus (\ker Z_Y)^\perp$, Z_Y is block diagonal with blocks 0 and $Z_Y^\perp := (1-\Pi_Y) Z_Y (1-\Pi_Y)$; the hypotheses give $\Pi_Y D \Pi_Y = 0$ and $(1-\Pi_Y) D \Pi_Y = 0$, hence also $\Pi_Y D (1-\Pi_Y) = ((1-\Pi_Y) D \Pi_Y)^\dagger = 0$. So $Z_Y - D$ is block diagonal with blocks 0 and $Z_Y^\perp - (1-\Pi_Y) D (1-\Pi_Y)$. On $(\ker Z_Y)^\perp$, $Z_Y^\perp \succeq \lambda_{\min}^+(Z_Y)$ and $(1-\Pi_Y) D (1-\Pi_Y) \preceq \|(1-\Pi_Y) D (1-\Pi_Y)\| \le \lambda_{\min}^+(Z_Y)$, so the second block is PSD. Thus $Z_Y - D \succeq 0$, and (a) gives $P_X = P_H$.

(e) Let $M_n(X) = \mathbb{E}_{x \in X} (x x^\dagger)^{\otimes n}$ and $M_n(H) = \int_H (hf\,(hf)^\dagger)^{\otimes n} dh$ in $\mathrm{End}(V^{\otimes n})$, and $m = M_n(X) - M_n(H)$. Let τ_T be the transpose on the first T tensor factors, $\tau_T(A_1 \otimes A_2) = A_1^{\mathrm{t}} \otimes A_2$ for $A_1 \in \mathrm{End}(V^{\otimes T})$, $A_2 \in \mathrm{End}(V^{\otimes (T+k)})$, extended linearly; we read its output as an operator on $\bar{V}^{\otimes T} \otimes V^{\otimes (T+k)}$ (same underlying space and basis). Let $P_{\mathcal{K}}$ be the orthogonal projection onto $\mathcal{K}$ and $\mathcal{L}(A) = P_{\mathcal{K}} \, \tau_T(A) \, P_{\mathcal{K}}$, viewed in $\mathrm{End}(\mathcal{K})$.

$D = \mathcal{L}(m)$. $(x x^\dagger)^{\mathrm{t}} = \bar{x} \bar{x}^\dagger$, so $\tau_T((x x^\dagger)^{\otimes n}) = u_x u_x^\dagger$, which already lives on $\mathcal{K}$; by linearity (and continuity, for the Haar integral) $\mathcal{L}(M_n(X)) = R_X$, $\mathcal{L}(M_n(H)) = R_H$, and $D = \mathcal{L}(m)$.

Equivariance. For product operators, $(h A_1 h^\dagger)^{\mathrm{t}} = \bar{h} \, A_1^{\mathrm{t}} \, h^{\mathrm{t}} = \bar{h} A_1^{\mathrm{t}} \bar{h}^\dagger$ (as $h^{\mathrm{t}} = \bar{h}^\dagger$), so $\tau_T(\mathrm{Ad}_h A) = (\bar{h}^{\otimes T} \otimes h^{\otimes (T+k)}) \, \tau_T(A) \, (\bar{h}^{\otimes T} \otimes h^{\otimes (T+k)})^\dagger$ for all A by linearity. $\mathcal{K}$ is invariant under the unitary $\bar{h}^{\otimes T} \otimes h^{\otimes (T+k)}$, so $P_{\mathcal{K}}$ commutes with it, and $\mathcal{L}(\mathrm{Ad}_h A) = U_h \mathcal{L}(A) U_h^\dagger$. For $Y \in \mathcal{Y}_H$ define $\Phi_Y : \mathrm{End}(V^{\otimes n}) \to \mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp)$, $\Phi_Y(A) = (1-\Pi_Y) \mathcal{L}(A)|_{\ker Z_Y}$. Since Π_Y commutes with U_h,

$$\Phi_Y(\mathrm{Ad}_h A) = (1-\Pi_Y) U_h \mathcal{L}(A) U_h^\dagger \Pi_Y = U_h|_{(\ker Z_Y)^\perp} \, \Phi_Y(A) \, U_h|_{\ker Z_Y}^\dagger ,$$

i.e. Φ_Y is H-equivariant for the actions (A3), and $(1-\Pi_Y) D \Pi_Y = \Phi_Y(m)$ (extended by 0 on $(\ker Z_Y)^\perp$).

Isotypic decomposition with multiplicities. H is compact, so the finite-dimensional modules $\mathcal{E} = \mathrm{End}(V^{\otimes n})$ and $\mathcal{H}_Y = \mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp) \cong (\ker Z_Y)^\perp \otimes (\ker Z_Y)^*$ decompose into isotypic components, $\mathcal{E} = \bigoplus_\lambda \mathcal{E}_\lambda$ with $\mathcal{E}_\lambda \cong V_\lambda \otimes \mathbb{C}^{\mu_\lambda}$, and similarly $\mathcal{H}_Y = \bigoplus_\lambda (\mathcal{H}_Y)_\lambda$; the isotypic projections P_λ are H-equivariant (they are $\int_H \overline{\chi_\lambda(h)} \dim V_\lambda \, \mathrm{Ad}_h \, dh$). By Schur's lemma, an equivariant map sends $\mathcal{E}_\lambda$ into $(\mathcal{H}_Y)_\lambda$, whatever the multiplicities: restricted to any irreducible summand $\cong V_\lambda$ of $\mathcal{E}_\lambda$, its components into the irreducible summands of $\mathcal{H}_Y$ vanish unless they are $\cong V_\lambda$. Hence $\Phi_Y(m_\lambda) = 0$ unless $V_\lambda \hookrightarrow \mathcal{H}_Y$, where $m_\lambda = P_\lambda m$. The dual $(\ker Z_Y)^*$ enters only through this Hom space; for self-dual H-modules (e.g. all $SU(2)$- and $SO(4)$-modules) $(\ker Z_Y)^* \cong \ker Z_Y$.

Trivial component. $P_{\mathbf{1}} = \int_H \mathrm{Ad}_h \, dh$, and $P_{\mathbf{1}} M_n(X) = M_n(H)$ by the argument of (3) (each

$x \in X$ lies in $H \cdot f$), while $P_{\mathbf{1}} M_n(H) = M_n(H)$; so $m_{\mathbf{1}} = 0$.

G-fixed vectors. $M_n(X)$ is Ad_G-invariant and $M_n(H)$ is Ad_H-invariant, so m is G-fixed; P_λ commutes with Ad_g, so m_λ is a G-fixed vector of $\mathcal{E}_\lambda \cong V_\lambda \otimes \mathbb{C}^{\mu_\lambda}$, whose G-fixed space is $V_\lambda^G \otimes \mathbb{C}^{\mu_\lambda}$. Hence $m_\lambda \neq 0$ requires $V_\lambda^G \neq 0$.

Therefore $(1 - \Pi_Y) D \Pi_Y = \sum_\lambda \Phi_Y(m_\lambda)$ only receives contributions from $\lambda \neq \mathbf{1}$ with $V_\lambda^G \neq 0$ and $V_\lambda \hookrightarrow \mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp)$. Since $m = M_{2T+k}(X) - M_{2T+k}(H)$, this is (e). □

Corollary 34 (W-SU(2))**.** *Let $H = SU(2)$ act on $V = V_j$ (spin j), $G \leq H$ closed, and let $Y \in \mathcal{Y}_H$ with $\ker Z_Y$ spin-s isotypic. Then $(1 - \Pi_Y) D \Pi_Y$ only receives contributions from spins L with $1 \leq L \leq \min\{2j(2T+k),\ s + j(2T+k)\}$ and $V_L^G \neq 0$.*

Proof. $V_j^{\otimes n}$ contains only spins $\leq jn$, so $\mathrm{End}(V_j^{\otimes n}) \cong V_j^{\otimes n} \otimes (V_j^{\otimes n})^*$ contains only spins $L \leq 2jn$ (every $SU(2)$-irreducible is self-dual). $\bar{V}_j \cong V_j^* \cong V_j$, so $\mathcal{K} \subset \bar{V}_j^{\otimes T} \otimes V_j^{\otimes (T+k)}$ contains only spins $p \leq jn$, and so does $(\ker Z_Y)^\perp$. Write $\ker Z_Y \cong V_s \otimes \mathbb{C}^\mu$ and $(\ker Z_Y)^\perp \cong \bigoplus_p V_p \otimes \mathbb{C}^{\nu_p}$; then $\mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp) \cong \bigoplus_p (V_p \otimes V_s^*) \otimes \mathbb{C}^{\mu\nu_p}$, and multiplicities create no new irreducibles. By the triangle rule, $V_L \hookrightarrow V_p \otimes V_s^* \cong V_p \otimes V_s$ requires $L \leq s + p \leq s + jn$. $L = 0$ is excluded by (e). Apply Theorem 32(e). (No use of (C1).) □

Corollary 35 (W-O(4))**.** *Let $H = O(4)$ act on $V = \mathbb{C}^4$ by its defining real orthogonal representation, let $f \in \mathbb{R}^4$ be a real unit vector and $G \leq O(4)$ finite, and suppose $X = G \cdot f \subset S^3$ is a spherical t-design. Let $Y \in \mathcal{Y}_H$ and suppose $\ker Z_Y$ is isotypic, as an $SO(4)$-module, of type (a, b) under $SO(4) \cong (SU(2) \times SU(2))/\{\pm 1\}$. Then $(1 - \Pi_Y) D \Pi_Y = \sum_\ell (1 - \Pi_Y) \mathcal{L}(m_\ell) \Pi_Y$, where the sum runs over even ℓ with*

$$t < \ell \leq 2(2T + k), \qquad \ell \leq 2\min(a, b) + 2T + k,$$

and m_ℓ is the degree-ℓ spherical-harmonic part of m defined in the proof. In particular, if no even ℓ satisfies both bounds, then $(1 - \Pi_Y) D \Pi_Y = 0$.

Proof. Step 1 (harmonic parts of m). For real x, $xx^\dagger = xx^{\mathrm{t}}$ and $\phi(x) := (xx^{\mathrm{t}})^{\otimes n}$ has entries that are homogeneous polynomials of degree $2n$ in x. On S^3 each entry decomposes into spherical harmonics of degrees $\ell \leq 2n$, ℓ even (the entries are even functions); write $\phi = \sum_\ell \phi_\ell$ entrywise. Since $O(4)$ commutes with the harmonic projections and $\phi(hx) = h^{\otimes n} \phi(x) h^{\mathrm{t} \otimes n}$, also $\phi_\ell(hx) = h^{\otimes n} \phi_\ell(x) h^{\mathrm{t} \otimes n}$. $O(4)$ acts transitively on S^3, so $M_n(H) = \int_{S^3} \phi \, d\sigma$, and $m = \sum_\ell m_\ell$ with $m_\ell := \mathbb{E}_X \phi_\ell - \int_{S^3} \phi_\ell \, d\sigma$. For $\ell \geq 1$, $\int \phi_\ell \, d\sigma = 0$; for $1 \leq \ell \leq t$, $\mathbb{E}_X \phi_\ell = 0$ by the design property; and $m_0 = 0$. So $m = \sum_{t < \ell \leq 2n,\ \ell \text{ even}} m_\ell$ with $m_\ell = \mathbb{E}_X \phi_\ell$. Let $\mathcal{H}_\ell$ be the space of degree-ℓ harmonics, $Z_y^{(\ell)} \in \mathcal{H}_\ell$ the reproducing kernel at y, and $\Psi_\ell(p) = \int_{S^3} p(x) \phi(x) \, d\sigma(x)$ for $p \in \mathcal{H}_\ell$. Ψ_ℓ is $SO(4)$-equivariant and $\phi_\ell(y) = \Psi_\ell(Z_y^{(\ell)})$, so $m_\ell = \Psi_\ell(\mathbb{E}_X Z_y^{(\ell)}) \in \Psi_\ell(\mathcal{H}_\ell)$. Since $\mathcal{H}_\ell \cong (\ell/2, \ell/2)$ is an irreducible $SO(4)$-module, m_ℓ lies in the $(\ell/2, \ell/2)$-isotypic component.

Step 2 (types in $\mathcal{K}$). $V \otimes_{\mathbb{R}} \mathbb{C}$ is $(\frac{1}{2}, \frac{1}{2})$, and $\mathrm{Sym}^i(\mathbb{C}^4) = \bigoplus_{i' \leq i,\ i' \equiv i\ (2)} \mathcal{H}_{i'} \cong \bigoplus (i'/2, i'/2)$. $\bar{V} \cong V$, so by Clebsch–Gordan in each $SU(2)$ factor $\mathcal{K} = \mathrm{Sym}^T(\bar{V}) \otimes \mathrm{Sym}^{T+k}(V)$ contains only types (p, p') with $p, p' \leq (T + T + k)/2 = n/2$; so does $(\ker Z_Y)^\perp \subset \mathcal{K}$.

Step 3 (the Hom space). The map Φ_Y of the proof of (e) is $O(4)$-equivariant, hence $SO(4)$-equivariant, so by the Schur argument of (e) (applied to $SO(4)$, with multiplicities) it kills m_ℓ unless $(\ell/2, \ell/2) \hookrightarrow \mathrm{Hom}(\ker Z_Y, (\ker Z_Y)^\perp) \cong (\ker Z_Y)^* \otimes (\ker Z_Y)^\perp$. $SU(2)$-irreducibles are self-dual, so $(a, b)^* \cong (a, b)$, and $(\ell/2, \ell/2) \hookrightarrow (a, b) \otimes (p, p')$ requires $\ell/2 \leq a + p$ and $\ell/2 \leq b + p'$. With $p, p' \leq n/2$ this gives $\ell \leq 2\min(a, b) + n$. □

Table 4: Finite-orbit tests at matched $(T+k)$-th moment (genuine cases: Lemma 5 is not tight).

H / fiducial	G-orbit	(T,k)	$P_X - P_H$	status
$SO(3)$, real $\|1,0\rangle$	6 icosahedral axes	(1,1)	$< 10^{-13}$	pass
$SO(3)$, real $\|1,0\rangle$	30-line icosahedral orbit	(2,1)	$< 10^{-12}$	pass
$SO(3)$, real $\|1,0\rangle$	30-line icosahedral orbit	(1,3)	$< 10^{-11}$	pass
$SU(2)$, spin 3/2 GHZ-like	icosahedral (tuned)	(1,1)	$< 10^{-12}$	pass ($P = 31/70$)
$O(4)$, real	24-cell	(1,1)	$< 10^{-8}$	pass
$SU(2)$, spin 1 (via $SO(3)$)	octahedral (tuned)	(1,1)	$\geq 6.437 \times 10^{-4}$	fail (certified)
$SU(2)$, spin 5/2	binary icosahedral (tuned)	(1,1)	$1.1\text{–}8.8 \times 10^{-4}$	fail (predicted by Thm. 32)
$SU(3)$, $\mathrm{Sym}^2\,\mathbb{C}^3$	Hessian (tuned)	(1,1)	$\geq 8.0399 \times 10^{-4}$	fail (certified)

Remark 36. Corollaries 34 and 35 only restrict where $(1-\Pi_Y)D\Pi_Y$ can be nonzero. To conclude $P_X = P_H$ one still needs (b) (or $\Pi_Y D\Pi_Y = 0$ directly) and the norm condition of (d). Applying them to a concrete orbit (e.g. the $O(4)$ row of Table 4) requires Y and the type of $\ker Z_Y$, which were obtained numerically [Num]. In the five $O(4)$ tests of the research phase the computed window degrees $4, 6, 8, 8, 8$ equal $2\min(a,b) + 2T + k$ and lie below the first surviving harmonic (6 for the F_4 orbit, 12 for the H_4 orbit).

8 Related work

Optimal cloning for continuous families. Werner [12] constructed the unique optimal $N \to M$ cloner for pure d-level states under global fidelity. For equatorial (phase) states, Bruß *et al.* [20] bounded the $N \to M$ fidelity via phase estimation and gave the optimal $1 \to 2$ qubit machine; Fan *et al.* [21] obtained the optimal $1 \to 2$ phase-covariant cloner for qudits under single-qudit fidelity; and Buscemi, D'Ariano and Macchiavello [7] derived the optimal $N \to M$ phase-covariant cloner for $M = kd + N$ (their k is an integer multiplier, not our number of extra copies), optimal for both global and single-qudit fidelity. Kim and Chitambar [8] computed the optimal global (process) fidelity of phase-covariant cloning by semidefinite programming, with closed form $(2d-1)/d^2$ for $1 \to 2$, and asked for a $1 \to M$ formula. That formula, and in fact the optimal $N \to N + K$ phase-state cloner for all d, N, K, is now given by Herasymenko, Lin, Verduyn Lunel and Grinko [9, Thm. II.3 and Thm. X.1, Eq. (16)]. In particular the $T = 1$ torus values, $(k+1)!(2d-k)/(2d^{k+1})$ for $k \leq d$, are the $N = 1$ case of their Eq. (16); we use their result for the torus value and for the singleton case of Proposition 11, and claim no novelty for it. Asymptotic and asymmetric variants are treated by Chiribella and Yang [18], who relate optimal asymptotic cloners to state estimation for, among others, phase- and multiphase-covariant families, and by Kay [22] for universal cloning. *Difference.* All of these works optimise over the full continuous orbit (the torus of phases, or the unitary group). We ask a different question: which *finite* phase codebooks $\{\psi_x\}_{x \in \mathbb{Z}_q}$ with key set $K \subset \mathbb{Z}_q$ attain the torus value P_{torus} exactly.

Finite sets that reproduce a continuous optimum. The closest precedent is Jing *et al.* [13]: three equally spaced equatorial qubit states determine the phase-covariant cloner, and four tetrahedral states determine the universal one. Their statement is for qubits and single-copy fidelity. For $d = 2$, $T = k = 1$, our sufficiency result gives the same conclusion under global fidelity, since $\{0,1\} \subset \mathbb{Z}_3$ is a B_2 set. Our necessity statement (Theorem 7: Sidon is required at $(T,k) = (1,1)$) holds *for global fidelity only* and does not transfer to single-copy fidelity, where Jing *et al.* report

that three states suffice for all $1 \to n$. For larger (T, k) we prove sufficiency of B_{T+k} only; necessity is open. Ćwikliński and Studziński [19] use moment matching in the opposite direction: when the second moment of an orbit equals the Haar moment, the orbit is a continuous 2-design and local cloning reduces to universal cloning. Herasymenko *et al.* [9] show that stabilizer states are not a resolving orbit by a 4-copy frame-potential computation; Gross, Nezami and Walter [23] supply the Clifford-group Schur–Weyl duality that controls such stabilizer moments. *Difference.* We give (i) an exact finite-codebook criterion on the torus: B_{T+k} index sets attain P_{torus}, proved for $d = 2$ and for $T \leq 3$ (all d, k), with an if-and-only-if (Sidon) characterisation at $(T, k) = (1, 1)$; and (ii) a window theorem (Theorem 32) for finite orbits of compact groups, with two computer-assisted counterexamples (SU(3) on $\mathrm{Sym}^2\mathbb{C}^3$ with the Hessian group, and spin-1 with the octahedral group) showing that the finite-subgroup analogue fails beyond tori. We found no prior finite-orbit failure theorem or window criterion.

Quantum fingerprinting and hashing. Buhrman, Cleve, Watrous and de Wolf [1] introduced quantum fingerprints that are exponentially shorter than the strings they encode; Gottesman and Chuang [6] built unconditionally secure quantum digital signatures from quantum public keys, whose security requires that only a limited number of copies circulate. Gavinsky and Ito [24] constructed *hiding* quantum fingerprints that reveal $o(\min\{n, \log(1/\varepsilon)\})$ bits. Ablayev and Vasiliev [2] introduced cryptographic quantum hashing with a Fourier-controlled key set and a trigonometric (qubit-rotation) encoding; Vasiliev [3] formulated the character-based phase encoding $\frac{1}{\sqrt{d}} \sum_j e^{2\pi i s_j x/q} |j\rangle$ for finite abelian groups, which is exactly our state family, with an ε-biased key set $S \subset \mathbb{Z}_q$ giving collision resistance (see also the review [5]); Ziiatdinov [4] extends the construction to arbitrary groups. Hatanaka *et al.* [25] propose a photonic hash based on Gaussian boson sampling. Adaptive attacks on Wiesner money [17] illustrate how copying and verification interact in practice. *Difference.* These works use Fourier bias, i.e. near-orthogonality, for collision resistance. None of them analyses optimal *cloning* of the key-set states, and none uses a Sidon or B_h condition. Our results give the corresponding copy-resistance statement: for B_{T+k} key sets, an adversary holding T copies does no better at producing $T + k$ copies (in global fidelity) than against the full phase torus.

Sidon sets, difference sets and their uses in quantum information. Sidon (B_2) sets go back to Erdős and Turán [10]. Singer [26] gave perfect difference sets modulo $q^2 + q + 1$, and Bose and Chowla [27] extended these results to B_h sequences. Katz's character-sum estimate [28] bounds $|\sum_{t \in F} \chi(t - x)| \leq (n-1)\sqrt{\#F}$ for x of degree n over F; it controls the Fourier bias of Bose–Chowla-type sets, which we use in the Discussion. O'Bryant's survey [11] is the standard bibliography. In quantum information, Bertoni, Mereghetti and Palano [29] used Golomb rulers and difference sets to minimise the size of one-way quantum finite automata, and Aydin, Albert and Barg [30] built ℓ_1 codes, and hence spin, bosonic and permutation-invariant quantum codes, from t-Sidon sets and Tverberg partitions. Vitelli *et al.* [31] realised quantum state fusion of two photonic qubits into one ququart. *Difference.* In [29, 30] the additive structure serves automaton size or code distance. Here it is the exact condition for finite codebooks to attain optimal cloning. Our identity between the optimal fusion fidelity of phase states and the additive energy of the key set (Theorem 26) has no counterpart in [31], which is a physical protocol. The multiset Johnson-type packing bound used in the proofs of Theorem 13(c,d) is a packing statement; it is not the Erdős–Ko–Rado bound for intersecting multiset families of Meagher and Purdy [32], which goes in the opposite direction.

9 Discussion

For quantum hashing and copy-bounded quantum tokens, our results replace the bias heuristic by an exact criterion: at $(T, k) = (1, 1)$ a phase codebook is as hard to counterfeit as the full torus exactly when its key set is Sidon, and for larger (T, k) the absence of additive coincidences of order $T + k$ (B_{T+k}) suffices (proved for $d = 2$ and for $T \leq 3$; conjectured in general); whether it is also necessary is open. Bose–Chowla sets [27] are B_h with small bias [28], but they are not efficiently indexable on qubit registers, and the scheme remains malleable and vulnerable to verification-oracle attacks; these results are information-theoretic statements about copy-bounded security, not a deployable primitive.

Open problems. (1) Conjecture 14 for $T \geq 4$; the certified margin $1 - \max \rho \approx 2.9/T^2$ at $d = 3$ shows that a proof must be sharp in T. (2) Exact conditions for $T \geq 2$: Theorem 18 settles $T = 1$ (and shows that B_{T+k} is not necessary); is there an analogous characterisation for $T \geq 2$, and does Conjecture 23 hold for all (d, k)? (3) A necessary-and-sufficient form of Theorem 32, i.e. a characterisation of the finite orbits whose cloning value equals that of the continuous orbit. (4) Codebooks over other finite abelian groups (e.g. $\mathbb{Z}_q^r$), and cloning figures of merit other than global fidelity.

Use of AI tools

This work made substantial use of four AI systems.

- **Grok Bot** (xAI's agent platform, launched on the Grok 4.6 model; used October 2026) was used throughout the research phase. It proposed and drafted proofs (the reduction to the packing inequality, the packing inequality for $d = 2$ and $T \leq 3$, the fusion theorems, the $T = 1$ characterisation and the window theorem), wrote exploratory numerical code and the certificate generators, wrote the standalone exact, interval and ball-arithmetic checkers of Appendix C, and ran literature and prior-art searches.

- **Claude** (Anthropic; Claude Opus 5.5) co-developed research directions and conjectures, independently re-implemented and audited every computational claim in a separate code base (QH-Lab [33]), re-ran every certificate checker, verified every bibliographic entry against Crossref and arXiv, and drafted and edited the manuscript.

- **ChatGPT** (OpenAI; GPT-6 Luna, the ChatGPT Free model since 8 October 2026; used 16 October 2026) served as an independent referee of the draft, re-ran the standard-library certificate checkers, and gave an independent proof of Theorem 28, and **Perplexity** (Perplexity Pro with the GPT-6.1 Sol model) as an independent prior-art and citation-fidelity checker. Every issue they raised was adjudicated by recomputation or by a written proof before any change was made.

The outputs were verified as follows. (1) Every computer-assisted step is checked by a standalone program that uses only exact rational, interval or ball arithmetic and no optimisation solver; floating-point and solver outputs serve only as certificate data. (2) The complete certificate package was re-run independently: the archive hash and all 32 internal hashes matched, and every output was identical to the shipped expected output. (3) The values in Tables 1–4 and in the negative-results tables, the worked examples, the m-fold fusion values and the $T = 1$ characterisation (on random

key sets) were reproduced by an independent implementation (QH-Lab), together with floating-point spot checks of the torus theorem at $T = 2$ ($d \leq 20$) and $T = 3$ ($d \leq 8$), which agree with the torus value to within 10^{-9}. (4) The pen-and-paper arguments were reviewed line by line by Claude against the research record, and by two independent ChatGPT referee passes on successive drafts; the author read the complete manuscript. (5) Every reference was checked against its arXiv or DOI landing page. The author takes full responsibility for the content of this paper, including all text and results produced with AI assistance.

Data and code availability

All code needed to verify the computer-assisted steps is provided as arXiv ancillary files and in the Zenodo record that accompanies this paper [33] (code under the MIT licence). The archive `ca_appendix.zip` has SHA-256 `e77e131bdd7becb0822ed582ccfc0a9e43b4b9183aeab37e65bec3e64d888b30`; the separate coverage audit `coverage_check.py` has SHA-256 `c42d11f652ed0bdd7daec55dd4bb9940319a8ff138c7f`. The values in the negative-results tables are reproduced by QH-Lab's `tools/neg_table_check.py`. Verification requires Python ≥ 3.10, `mpmath` and `python-flint` 0.9.0. No other data were generated or analysed.

Author contributions

M. Weldon is the sole author. He conceived the project, directed the research and its multi-system verification workflow, read and approved the final manuscript, and takes responsibility for its content, with AI assistance as described above.

Competing interests

The author declares no competing interests. This work received no external funding.

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A Proof of the packing inequality (Theorem 13)

Notation and scope. Throughout, $N = T + k$. A *shift* is $\Delta \in \mathbb{Z}^d$ with $\sum_i \Delta_i = k$; $\Delta = \Delta^+ - \Delta^-$ with $\Delta^\pm \geq 0$ of disjoint supports, and $n(\Delta) = |\Delta^-|$. The *support* of Δ is $\mathcal{D}_\Delta = \{b : |b| = T,\ b \geq \Delta^-\}$ (this is $\mathcal{D}_\Delta$ of [9, Eq. (111)]) and its *outputs* are $b + \Delta$, $b \in \mathcal{D}_\Delta$. A *family* is a finite set F of shifts with $n(\Delta) \leq T$ for all $\Delta \in F$ that pairwise satisfy

$$\text{(ii)} \qquad \|\Delta - \Delta'\|_1 \geq 2(T + k + 1).$$

For $x, y \in \mathbb{Z}^d_{\geq 0}$: $|x| = \sum_i x_i$, $\mathrm{ov}(x, y) = \sum_i \min(x_i, y_i)$, $\mathrm{mult}(x) = |x|!/\prod_i x_i!$. $\mathrm{Mb}(M, \ell)$ is the largest multinomial $\mathrm{mult}(x)$ over x with $|x| = M$ and at most ℓ nonzero coordinates (attained by the balanced vector), $M^* = \mathrm{Mb}(N, d)$, and $R_m = m!/m^m$. We write $c_b = \mathrm{mult}(b)$ for $|b| = T$, so $a_b = \sqrt{c_b/Z_b}$. The tag [CA: `path`] marks a step certified by the named standalone checker of Appendix C (paths relative to `ca_appendix/`); every other step is proved in the text.

Remark 37 (Scope of "family"). A shift with $n(\Delta) > T$ has $\mathcal{D}_\Delta = \emptyset$ and $\Lambda(\Delta) = 0$, and it can be at distance $\geq 2(T+k+1)$ from a balanced g; e.g. $d = 3$, $T = k = 1$, $g = (1,0,0)$, $\Delta' = (-1,-1,3)$, $\|g - \Delta'\|_1 = 6$. Then $\rho(\{g, \Delta'\}) = 1$. So parts (b)–(d) of Theorem 13 (and the strict inequality of Conjecture 14) hold only for families with $n(\Delta) \leq T$, which is how they are proved here; for Proposition 11 the restriction is harmless, since members with $\Lambda = 0$ do not change ρ. Part (a) is proved below without the restriction.

A.1 Common lemmas

Lemma 38 (Normalisation, Lemma N). *Let* $\mathrm{avg}_b = |C|^{-1} \sum_{g \in C} \mathrm{mult}(b+g)$, $\theta_b = \sqrt{c_b/\mathrm{avg}_b}$ *and* $W = \sum_{|b|=T} \sqrt{c_b \,\mathrm{avg}_b}$. *Then for every finite set* F *of shifts*

$$\rho(F) = \frac{1}{W} \sum_{\Delta \in F} \sum_{b \in \mathcal{D}_\Delta} \theta_b \,\mathrm{mult}(b+\Delta).$$

Proof. $Z_b = |C| \,\mathrm{avg}_b$, so $a_b = \theta_b/\sqrt{|C|}$. The a_b are S_d-invariant and C is one S_d-orbit, so $\Lambda(g) = \Lambda_0$ for all $g \in C$; averaging over C, $\Lambda_0 = \sum_b a_b \,\mathrm{avg}_b = |C|^{-1/2} \sum_b \theta_b \mathrm{avg}_b = W/\sqrt{|C|}$. Divide $\sum_F \Lambda(\Delta) = |C|^{-1/2} \sum_F \sum_b \theta_b \,\mathrm{mult}(b+\Delta)$ by Λ_0. □

Write $\kappa_b = \mathrm{avg}_b/M^*$. Then $\theta_b \,\mathrm{mult}(\cdot)/W = \sqrt{c_b/\kappa_b}\,(\mathrm{mult}(\cdot)/M^*)/\sum_b \sqrt{c_b \kappa_b}$, so all bounds below may be computed in units of M^* with $\theta_b = \sqrt{c_b/\kappa_b}$ and $W = \sum_b \sqrt{c_b \kappa_b}$.

Lemma 39 (Structure, Lemma S). *For shifts* Δ, Δ' *with* $n = n(\Delta)$, $n' = n(\Delta')$: $\|\Delta\|_1 = k + 2n$, *and* $\|\Delta - \Delta'\|_1 = \|\Delta\|_1 + \|\Delta'\|_1 - 2\mathrm{ov}(\Delta^+, \Delta'^+) - 2\mathrm{ov}(\Delta^-, \Delta'^-)$. *Hence*

$$\textit{(ii)} \iff \mathrm{ov}(\Delta^+, \Delta'^+) + \mathrm{ov}(\Delta^-, \Delta'^-) \leq n + n' - T - 1.$$

Consequently, in a family: (S1) every pair has $n + n' \geq T + 1$; *(S2) at most one member has* $n \leq T/2$; *(S3) if* $|F| \geq 2$, *every member has* $n \geq 1$; *(S4) a member with* $n = T$ *is* pinned: $\mathcal{D}_\Delta = \{\Delta^-\}$; *(S5) supports of distinct members are disjoint and outputs of distinct members are distinct.*

Proof. $|\Delta^+| - |\Delta^-| = k$ gives $\|\Delta\|_1 = |\Delta^+| + |\Delta^-| = k + 2n$. For integers $x = x^+ - x^-$, $y = y^+ - y^-$ with $x^+x^- = y^+y^- = 0$ one checks $|x - y| = |x| + |y| - 2\min(x^+, y^+) - 2\min(x^-, y^-)$ (if $x, y \geq 0$ this is $|x-y| = x+y-2\min(x,y)$; if $x \geq 0 \geq y$ both minima vanish; the other cases are symmetric). Summing over coordinates gives the identity, and substituting into (ii) the equivalence. (S1): the left side is ≥ 0. (S2): two members with $n, n' \leq T/2$ would have $n + n' \leq T$. (S3): by (S1) a partner of a member with $n = 0$ has $n' \geq T + 1$, which is excluded in a family. (S4): $b \geq \Delta^-$ and $|b| = T = |\Delta^-|$ force $b = \Delta^-$. (S5): if $b \in \mathcal{D}_\Delta \cap \mathcal{D}_{\Delta'}$, then $\beta = b + \Delta$, $\beta' = b + \Delta'$ are nonnegative with $|\beta| = |\beta'| = N$, so $\|\Delta - \Delta'\|_1 = \|\beta - \beta'\|_1 \leq 2N$, contradicting (ii); if $b + \Delta = b' + \Delta'$, then $\|\Delta - \Delta'\|_1 = \|b' - b\|_1 \leq 2T$, again a contradiction. □

Lemma 40 (Multinomial moves, Lemma M). *Moving one unit from a coordinate of value* a *to one of value* c *multiplies* mult *by* $a/(c+1)$. *Consequently:*

(M1) (Doubling) If $M > \ell$ *and* $\ell < d$, *then* $\mathrm{Mb}(M, \ell+1) \geq 2\mathrm{Mb}(M, \ell)$. *Hence* $\mathrm{Mb}(N, \ell) \leq M^* 2^{\ell - d}$ *for* $\ell \leq d \leq N$, *and* $\mathrm{Mb}(M, \ell) \leq M!\, 2^{\ell - M}$ *for* $1 \leq \ell \leq M$.

(M2) (Filling) Let $m = \lfloor N/d \rfloor$ *and let* $|x| = N$ *with some coordinate* $x_z = t_0 \leq m$. *Then* $\mathrm{mult}(x)/M^* \leq m!/(t_0!\, m^{m - t_0})$. *In particular* $\mathrm{Mb}(N, \ell)/M^* \leq R_m^{d-\ell}$ *for* $\ell \leq d$.

(M3) (f*-monotonicity) For* $1 \leq s < M$ *and* $s < d$: $\mathrm{Mb}(M, s+1)/(s+1) \geq \frac{2s}{s+1} \mathrm{Mb}(M, s)/s \geq \mathrm{Mb}(M, s)/s$.

Proof. The move rule is immediate from the factorials. (M1) A balanced vector with ℓ parts summing to $M > \ell$ has a part $a \geq 2$; moving one unit to an empty coordinate multiplies mult by $a \geq 2$ and gives a vector with $\ell+1$ parts. Iterating from ℓ to d (allowed since $N \geq d$) gives the first bound. For the second, a vector with $\ell \leq M$ positive parts x_i has $\prod x_i! \geq \prod 2^{x_i-1} = 2^{M-\ell}$. (M2) While $x_z = t < m$, the other $d-1$ coordinates sum to $N-t > N-m \geq (d-1)m$, so their maximum is $\geq m+1$; moving one unit from it to z multiplies mult by $\geq (m+1)/(t+1) \geq m/(t+1)$. After $m - t_0$ moves the vector is still of size N, so mult $\leq M^*$, i.e. $\mathrm{mult}(x) \prod_{t=t_0}^{m-1} \frac{m}{t+1} \leq M^*$, which is the claim. For the second statement, fill the $d-\ell$ empty coordinates one after another ($t_0 = 0$ each time; already filled coordinates have value m and are never the donor, since the donor has value $\geq m+1$). (M3) By (M1), $\mathrm{Mb}(M, s+1)/(s+1) \geq 2\mathrm{Mb}(M,s)/(s+1)$, and $2s \geq s+1$. □

Lemma 41 (Johnson-type multiset packing, Lemma J)**.** *Let $s_r(P)$ be the number of distinct r-sub-multisets of a multiset P on $[d]$. If the members of a collection of multisets pairwise satisfy $\mathrm{ov}(P,P') \leq r-1$, then $\sum_P s_r(P) \leq \binom{d+r-1}{r}$. For $|P| = M$ with support size ℓ and $\ell_j = \#\{x : P_x \geq j\}$:*

$$s_2(P) = \tbinom{\ell}{2} + \ell_2 \geq \tbinom{\ell}{2} + [M > \ell],$$
$$s_3(P) = \tbinom{\ell}{3} + \ell_2(\ell-1) + \ell_3 \geq \tbinom{\ell}{3} + (\ell-1)[M > \ell] + [M \geq \ell+2] =: s_3^{\min}(\ell, M).$$

Proof. An r-multiset below both P and P' would give $\mathrm{ov}(P,P') \geq r$; so the sets of r-sub-multisets are pairwise disjoint subsets of the $\binom{d+r-1}{r}$ r-multisets on $[d]$. The formulas count sub-multisets by shape ($\{x,y\}$, $\{x,x\}$; $\{x,y,z\}$, $\{x,x,y\}$, $\{x,x,x\}$). $M > \ell$ forces $\ell_2 \geq 1$. If $M \geq \ell+2$, either some coordinate is ≥ 3 ($\ell_3 \geq 1$, $\ell_2 \geq 1$) or two coordinates are ≥ 2 ($\ell_2 \geq 2$ and $2(\ell-1) \geq (\ell-1)+1$ since $\ell \geq 2$). □

Lemma 41 is the multiset analogue of the Johnson packing bound; it is not an Erdős–Ko–Rado statement [32].

A.2 Part (a): $d = 2$, all T, k

Throughout this subsection $d = 2$ and $N = T + k$. Write $b = (b, T-b)$ with $b \in \{0, \ldots, T\}$, $\Delta = (\delta, k-\delta)$ with $\delta \in \mathbb{Z}$, and $c_m = \binom{N}{m}$ for $0 \leq m \leq N$, $c_m = 0$ otherwise. The output $b + \Delta = (b+\delta, N-b-\delta)$ is nonnegative iff $0 \leq b+\delta \leq N$, so

$$\mathcal{D}_\Delta = \{b \in \mathbb{Z} : \ \max(0, -\delta) \leq b \leq \min(T, N-\delta)\} \quad \text{(an interval, possibly empty)},$$
$$\Lambda(\Delta) = \sum_{b=0}^{T} a_b \, c_{b+\delta}.$$

The boundary cases $b = 0$, $b = T$ and partial or empty supports are all covered by the convention $c_m = 0$ for $m \notin [0, N]$.

Lemma 42 (Dominated rearrangement)**.** *Let $x_0 \geq x_1 \geq \cdots \geq x_n \geq 0$ and let $(y_m)_{m \in Y}$ be nonnegative numbers on a finite index set Y with $|Y| \geq n+1$; let $y_{(0)} \geq \cdots \geq y_{(n)}$ be the $n+1$ largest of them (with multiplicity). If $U \subset \{0, \ldots, n\}$ and $\sigma : U \to Y$ is injective, then $\sum_{u \in U} x_u \, y_{\sigma(u)} \leq \sum_{i=0}^{n} x_i \, y_{(i)}$.*

Proof. Let $\gamma_0 \geq \gamma_1 \geq \ldots$ be the values $y_{\sigma(u)}$, $u \in U$, sorted decreasingly. They sit at $|U|$ distinct positions of Y (injectivity), so for each i the $i+1$ values $\gamma_0, \ldots, \gamma_i$ occupy $i+1$ distinct positions; at most i positions of Y carry a value $> y_{(i)}$, hence $\gamma_i \leq y_{(i)}$. Likewise the sorted values $\xi_0 \geq \xi_1 \geq \ldots$ of $(x_u)_{u \in U}$ satisfy $\xi_i \leq x_i$. By the rearrangement inequality and nonnegativity, $\sum_{u \in U} x_u y_{\sigma(u)} \leq \sum_i \xi_i \gamma_i \leq \sum_{i < |U|} x_i y_{(i)} \leq \sum_{i=0}^{n} x_i y_{(i)}$. □

Proof of Theorem 13(a). Let F be any finite set of shifts that pairwise satisfy (ii) (no condition on $n(\Delta)$ is needed when $d = 2$; members with empty support contribute 0).

Step 1: separation gives a partial injection. For $d = 2$, $\|\Delta - \Delta'\|_1 = |\delta - \delta'| + |(k - \delta) - (k - \delta')| = 2|\delta - \delta'|$, so (ii) reads $|\delta - \delta'| \geq N + 1$. (i) Supports are disjoint: if $b \in \mathcal{D}_\Delta \cap \mathcal{D}_{\Delta'}$, then $b + \delta, b + \delta' \in [0, N]$, so $|\delta - \delta'| \leq N$. (ii) Outputs are distinct: if $b + \delta = b' + \delta'$ with $b, b' \in [0, T]$, then $|\delta - \delta'| = |b - b'| \leq T < N + 1$. Let $U = \bigcup_{\Delta \in F} \mathcal{D}_\Delta \subset \{0, \ldots, T\}$; by (i) each $b \in U$ lies in the support of exactly one member $\Delta(b) = (\delta(b), k - \delta(b))$, and $\sigma(b) := b + \delta(b) \in [0, N]$ is injective on U by (ii). Hence

$$\sum_{\Delta \in F} \Lambda(\Delta) = \sum_{b \in U} a_b \, c_{\sigma(b)}.$$

Step 2: the $T + 1$ largest c-values, before any rearrangement. Let $c_{(0)} \geq \cdots \geq c_{(T)}$ be the $T + 1$ largest entries of $(c_m)_{m=0}^N$ ($N + 1 \geq T + 1$ entries). Put $\delta_0 = \lceil k/2 \rceil$ and $g = (\delta_0, k - \delta_0)$; $g \in C$ in both parities ($g = (k/2, k/2)$ for k even, $g = (\frac{k+1}{2}, \frac{k-1}{2})$ for k odd), and $\mathcal{D}_g = \{0, \ldots, T\}$ because $g \geq 0$. The window $\{\delta_0, \ldots, \delta_0 + T\}$ of row N is $\{\lceil k/2 \rceil, \ldots, N - \lfloor k/2 \rfloor\}$. Every window position i has $i \geq \lceil k/2 \rceil \geq \lfloor k/2 \rfloor$ and $N - i \geq \lfloor k/2 \rfloor$, so $\min(i, N - i) \geq \lfloor k/2 \rfloor$. Every omitted position j has either $j \leq \lceil k/2 \rceil - 1 \leq \lfloor k/2 \rfloor$ or $N - j \leq \lfloor k/2 \rfloor - 1$, so $\min(j, N - j) \leq \lfloor k/2 \rfloor$. Since $\binom{N}{m}$ is a nondecreasing function of $\min(m, N - m)$, every window value is $\geq$ every omitted value; hence the multiset $\{c_{b+\delta_0}\}_{b=0}^T$ is exactly $\{c_{(0)}, \ldots, c_{(T)}\}$.

Step 3: upper bound. By Lemma 42 with $x = (a_b)$ sorted decreasingly, $Y = \{0, \ldots, N\}$, $y = c$ and the injection σ of Step 1, $\sum_F \Lambda \leq \Lambda^* := \sum_{i=0}^T a_{(i)} c_{(i)}$, where $a_{(0)} \geq \cdots \geq a_{(T)}$ is the decreasing rearrangement of $(a_b)_{b=0}^T$.

Step 4: unimodality of a_b (all b, including $b = 0, T$). For even $L \geq 2$ and $0 \leq b \leq T$ put $r_L(b) = \binom{T}{b} / \binom{T+L}{b+L/2}$ (the lower index $b + L/2 \in [L/2, T + L/2]$ is always admissible). Then $r_L(T - b) = r_L(b)$ and, for $0 \leq b \leq T - 1$,

$$\frac{r_L(b+1)}{r_L(b)} = \frac{(T-b)(b+L/2+1)}{(b+1)(T+L/2-b)}, \qquad (T-b)(b+\tfrac{L}{2}+1) - (b+1)(T+\tfrac{L}{2}-b) = \tfrac{L}{2}(T-1-2b),$$

so r_L is nondecreasing on $0 \leq b \leq (T - 1)/2$; with the symmetry, $r_L(b)$ is a nonincreasing function of $|b - T/2|$. If k is even, $C = \{(k/2, k/2)\}$, $Z_b = \binom{N}{b+k/2}$ and $a_b^2 = r_k(b)$. If k is odd, $C = \{(\frac{k+1}{2}, \frac{k-1}{2}), (\frac{k-1}{2}, \frac{k+1}{2})\}$ and Pascal's rule gives $Z_b = \binom{N}{b+(k+1)/2} + \binom{N}{b+(k-1)/2} = \binom{N+1}{b+(k+1)/2}$, so $a_b^2 = r_{k+1}(b)$ (with $T + k + 1 = N + 1$). In both cases a_b is a nonincreasing function of $|b - T/2|$.

Step 5: $\Lambda_0 = \Lambda^*$. $\Lambda_0 = \Lambda(g) = \sum_{b=0}^T a_b c_{b+\delta_0}$. By Step 2 the second factors are $c_{(0)}, \ldots, c_{(T)}$ in some order, so it suffices that the pairing $b \mapsto (a_b, c_{b+\delta_0})$ is similarly ordered, i.e. that $a_b > a_{b'}$ implies $c_{b+\delta_0} \geq c_{b'+\delta_0}$ (then sorting by a with ties broken by c sorts both sequences, and $\Lambda_0 = \Lambda^*$). Note $c_{b+\delta_0}$ is a nonincreasing function of $|b + \delta_0 - N/2|$, which is $|b - T/2|$ for k even and $|b - \frac{T-1}{2}|$ for k odd. If $a_b > a_{b'}$, then $|b - \frac{T}{2}| < |b' - \frac{T}{2}|$ (Step 4). For k even this is the claim. For k odd, both numbers lie in $\frac{T}{2} + \mathbb{Z}$, so $|b - \frac{T}{2}| \leq |b' - \frac{T}{2}| - 1$, hence $|b - \frac{T-1}{2}| \leq |b - \frac{T}{2}| + \frac{1}{2} \leq |b' - \frac{T}{2}| - \frac{1}{2} \leq |b' - \frac{T-1}{2}|$, and $c_{b+\delta_0} \geq c_{b'+\delta_0}$.

Combining Steps 1–5, $\sum_F \Lambda \leq \Lambda^* = \Lambda_0$, i.e. $\rho(F) \leq 1$, with equality for $F = \{g\}$. □

A.3 Part (b): $T = 1$, all $d \geq 2$, $k \geq 1$

Here $N = k + 1$ and $|b| = 1$, so $b = e_a$. We prove the slightly stronger bound $\rho(F) \leq \frac{q+2}{2(d-1)(q+1)}$ for $k \geq d$, $q = \lfloor k/d \rfloor$, which implies the stated $3/(4(d - 1))$ because $q \geq 1$.

Proof of Theorem 13(b). Let F be a family (Remark 37) with $|F| \geq 2$.

Step 1: shape of the members. Every member has $n(\Delta) \in \{0,1\}$. A member with $n = 0$ and any partner with $n' \le 1$ would need, by Lemma 39, $\operatorname{ov}(\Delta^+, \Delta'^+) + \operatorname{ov}(\Delta^-, \Delta'^-) \le n + n' - T - 1 \le -1$, impossible. So every member has $n = 1$: $\Delta^- = e_a$ for one index a, and $\Delta^+ =: m_a$ satisfies $|m_a| = k + 1 = N$ and $(m_a)_a = 0$ ($\Delta^\pm$ have disjoint supports). We write $\Delta_a = m_a - e_a$.

Step 2: value of a member. $b \ge \Delta_a^- = e_a$ and $|b| = 1$ force $b = e_a$, so $\mathcal{D}_{\Delta_a} = \{e_a\}$ with output m_a, and $\Lambda(\Delta_a) = a_{e_a} \operatorname{mult}(m_a)$. Since C is S_d-invariant, Z_{e_a} and hence $a_1 := a_{e_a}$ do not depend on a.

Step 3: support disjointness. For two members $\Delta_a, \Delta_{a'}$, Lemma 39 gives

$$\|\Delta_a - \Delta_{a'}\|_1 = 2(k+2) - 2\operatorname{ov}(m_a, m_{a'}) - 2\operatorname{ov}(e_a, e_{a'}),$$

so (ii), i.e. $\|\Delta_a - \Delta_{a'}\|_1 \ge 2(k+2)$, holds iff $\operatorname{ov}(m_a, m_{a'}) = 0$ and $\operatorname{ov}(e_a, e_{a'}) = 0$. Thus the indices a of distinct members are distinct, and the supports $S_a = \operatorname{supp} m_a$ are pairwise disjoint subsets of $[d]$. Put $s_a = |S_a|$. Then $1 \le s_a \le s^* := \min(N, d-1)$ ($m_a \ne 0$; m_a has at most N positive entries; $S_a \subset [d] \setminus \{a\}$), and $\sum_a s_a \le d$.

Step 4: monotonicity of $f(s) = \operatorname{Mb}(N,s)/s$. For integers $1 \le s$ with $s + 1 \le \min(N, d)$, Lemma 40(M1) (applicable since $N > s$ and $s < d$) gives

$$\frac{f(s+1)}{f(s)} = \frac{s}{s+1} \cdot \frac{\operatorname{Mb}(N, s+1)}{\operatorname{Mb}(N,s)} \;\ge\; \frac{2s}{s+1} \;\ge\; 1.$$

So f is nondecreasing on $\{1, \dots, \min(N,d)\}$, and $s^* \le \min(N,d)$.

Step 5: numerator. $\operatorname{mult}(m_a) \le \operatorname{Mb}(N, s_a) = s_a f(s_a) \le s_a f(s^*)$, so

$$\textstyle\sum_a \Lambda(\Delta_a) \le a_1 f(s^*) \sum_a s_a \le a_1\, d\, f(s^*).$$

Step 6: denominator. Write $k = qd + s$ with $0 \le s \le d-1$; $g \in C$ has s coordinates $q+1$ and $d - s \ge 1$ coordinates q, and $\mathcal{D}_g = \{e_j\}_{j \in [d]}$. If $g_j = q$, then $g + e_j$ is balanced of size N, so $\operatorname{mult}(g + e_j) = \operatorname{Mb}(N,d)$. If $g_j = q+1$, moving one unit from coordinate j (value $q+2$) to a coordinate of value q yields a balanced vector and multiplies mult by $(q+2)/(q+1)$ (move rule), so $\operatorname{mult}(g + e_j) = \operatorname{Mb}(N,d)\frac{q+1}{q+2}$. Hence

$$\Lambda_0 = a_1 \sum_j \operatorname{mult}(g + e_j) = a_1 \operatorname{Mb}(N,d)\Big[(d-s) + s\,\frac{q+1}{q+2}\Big], \qquad \rho(F) \le \frac{d\, f(s^*)}{\operatorname{Mb}(N,d)\big[(d-s) + s\frac{q+1}{q+2}\big]}.$$

Case A: $1 \le k \le d-1$. Then $q = 0$, $s = k$, $N = k+1 \le d$, so $\operatorname{Mb}(N,d) = N!$ and the bracket is $d - k/2$. Here $s^* = N$ if $k \le d-2$ and $s^* = N-1$ if $k = d-1$; in both subcases $s^* \le N = \min(N,d)$, so $f(s^*) \le f(N) = N!/N = (N-1)!$ by Step 4. Hence

$$\rho(F) \le \frac{d\,(N-1)!}{N!\,(d - k/2)} = \frac{d}{(k+1)(d-k/2)},$$

and $(k+1)(d - \frac{k}{2}) - d = k\big(d - \frac{k+1}{2}\big) > 0$ (as $k + 1 \le d$), so the bound is < 1.

Case B: $k \ge d$. Then $q \ge 1$ and $N = k+1 \ge d+1$, so $s^* = d-1$ and $f(s^*) = \operatorname{Mb}(N, d-1)/(d-1)$. The bracket is $\ge d\frac{q+1}{q+2}$, and (M1) with $M = N > d-1 = \ell$, $\ell < d$ gives $\operatorname{Mb}(N,d) \ge 2\operatorname{Mb}(N, d-1)$. Hence

$$\rho(F) \le \frac{d\operatorname{Mb}(N,d-1)/(d-1)}{2\operatorname{Mb}(N,d-1)\, d\,\frac{q+1}{q+2}} = \frac{q+2}{2(d-1)(q+1)} \le \frac{3}{4(d-1)} < 1.$$

Cases A and B exhaust $k \ge 1$ for every $d \ge 2$ (for $d = 2$: Case A is $k = 1$, Case B is $k \ge 2$). □

For $k = 1$ the Case A bound is $d/(2d-1)$; at $d = 3$ it equals 0.6, the exact maximum found by the clique LP (research-phase computation [Num]), so the bound is sharp there.

A.4 Part (c): $T = 2$, all $d \geq 3$, $k \geq 1$

Here $N = k + 2$. The types of $|b| = 2$ are A ($b = 2e_i$, $c_b = 1$, d of them) and B ($b = e_i + e_j$, $c_b = 2$, $\binom{d}{2}$ of them); by symmetry κ_b and θ_b depend only on the type, and in units of M^*, $W = d\sqrt{\kappa_A} + \binom{d}{2}\sqrt{2\kappa_B}$, $\theta_A = 1/\sqrt{\kappa_A}$, $\theta_B = \sqrt{2/\kappa_B}$.

T2.1 Structure. By Lemma 39, $n \in \{1, 2\}$ (S3), at most one member has $n = 1$ (S2), and members with $n = 2$ are pinned (S4). The *light* member (if any) is $\Delta_1 = Q - e_i$ with $|Q| = k + 1$, $Q_i = 0$; its support is $\{2e_i\} \cup \{e_i + e_j\}_{j \neq i}$ with outputs $Q + e_i$ and $Q + e_j$. A *pinned* member is $\Delta = P - b$ with $|b| = 2$, $|P| = N$, $\mathrm{supp}P \cap \mathrm{supp}b = \emptyset$, and value $\theta_b \,\mathrm{mult}(P)$. The budgets of Lemma 39 give: pinned–pinned ($n + n' - 3 = 1$): $\mathrm{ov}(P, P') + \mathrm{ov}(b, b') \leq 1$, so the b's are distinct ($\mathrm{ov}(b, b) = 2$) and $\mathrm{ov}(P, P') \leq 1$; light–pinned ($n + n' - 3 = 0$): $\mathrm{ov}(P, Q) = 0$ and $b_i = 0$.

T2.2 Light value. With $t = |\mathrm{supp}Q|$ and the move rule $\mathrm{mult}(Q + e_x) = N\,\mathrm{mult}(Q)/(Q_x + 1)$,

$$\begin{aligned} V_1(Q) &= \theta_A \,\mathrm{mult}(Q + e_i) + \theta_B \sum_{j \neq i} \mathrm{mult}(Q + e_j) \\ &= N\,\mathrm{mult}(Q)\Big[\theta_A + \theta_B\Big((d - 1 - t) + \sum_{x \in \mathrm{supp}Q} \frac{1}{Q_x + 1}\Big)\Big] \\ &\leq \theta_A N \mathrm{Mb}(N - 1, d - 1) + (d - 1)\theta_B \mathrm{Mb}(N, d - 1). \end{aligned}$$

T2.3 Master bound. Drop the light–pinned constraints. At most d pinned members have type A, each of value $\leq \theta_A \mathrm{Mb}(N, d - 1)$. For type B, $\ell = |\mathrm{supp}P| \leq \min(d - 2, N)$, so by Lemma 41 ($r = 2$), $\mathrm{mult}(P) \leq \mathrm{Mb}(N, \ell) \leq \varphi_B\, s_2(P)$ with $\varphi_B = \max_{1 \leq \ell \leq \min(d-2,N)} \mathrm{Mb}(N, \ell)/(\binom{\ell}{2} + [N > \ell])$, and $\sum s_2(P) \leq \binom{d+1}{2}$. Hence

$$\rho(F) \leq S(d, k) := \frac{1}{W}\Big[\theta_A N \mathrm{Mb}(N-1, d-1) + (d-1)\theta_B \mathrm{Mb}(N, d-1) + \theta_B \varphi_B \tbinom{d+1}{2} + d\,\theta_A \mathrm{Mb}(N, d-1)\Big].$$

With $W \geq \binom{d}{2}\sqrt{2\kappa_B}$, $r_0 = \mathrm{Mb}(N, d - 1)/M^*$, $r_1 = N\mathrm{Mb}(N - 1, d - 1)/M^*$, $r_\varphi = \varphi_B/M^*$:

$$S \leq \frac{2}{d}\frac{r_0}{\kappa_B} + \frac{d+1}{d-1}\frac{r_\varphi}{\kappa_B} + \frac{2}{d-1}\frac{r_0}{\sqrt{2\kappa_A\kappa_B}} + \frac{2}{d(d-1)}\frac{r_1}{\sqrt{2\kappa_A\kappa_B}}. \tag{4}$$

T2.4 Ratio lemmas. Let $q_0 = \lfloor k/d \rfloor$.

(K) $\kappa_B \geq (\frac{q_0+1}{q_0+2})^2 \geq \frac{1}{4}$ and $\kappa_A \geq \frac{(q_0+1)^2}{(q_0+2)(q_0+3)} \geq \frac{1}{6}$. *Proof.* Fix $g \in C$ and let y be g plus two units placed on smallest coordinates; y is balanced of size N, so $\mathrm{mult}(y) = M^*$ and $\prod y_i! / \prod g_i! \geq (q_0 + 1)^2$. Also $\prod (b + g)_i! / \prod g_i! \leq (q_0 + 2)^2$ (type B) and $\leq (q_0 + 2)(q_0 + 3)$ (type A). Since $\mathrm{mult}(b + g)/M^* = \frac{\prod y_i! / \prod g_i!}{\prod (b+g)_i! / \prod g_i!}$, the bounds hold termwise in the average.

(K0) If $k \leq d - 2$: $M^* = N!$, g is a 0/1 vector with k ones, $\kappa_B = \mathbb{E}\, 2^{-X} \geq 1 - \mathbb{E}X/2 = 1 - k/d$ with $X = g_i + g_j$ (as $2^{-x} \geq 1 - x/2$ on $\{0, 1, 2\}$), $\kappa_A = \mathbb{E}[1/(2 + g_i)!] = \frac{1}{2} - \frac{k}{3d}$, and $r_\varphi \leq 1/\binom{N}{2}$. *Proof of the last bound.* If $N \leq d - 2$, $\ell = N$ gives $\mathrm{Mb}(N, N)/\binom{N}{2} = N!/\binom{N}{2}$. For $\ell \leq N - 1$, (M1) gives $\mathrm{Mb}(N, \ell)/(\binom{\ell}{2} + 1) \leq N!/g(\ell)$ with $g(\ell) = 2^{N-\ell}(\binom{\ell}{2} + 1)$; $g(\ell)/g(\ell + 1) = 2(\ell^2 - \ell + 2)/(\ell^2 + \ell + 2) \geq 1 \iff (\ell - 1)(\ell - 2) \geq 0$, so g is nonincreasing, and $g(N - 1) - \binom{N}{2} = \frac{1}{2}(N^2 - 5N + 8) > 0$.

(Moves) By (M2) with $m = \lfloor N/d \rfloor$: $r_0 \le R_m$ (one empty coordinate), $r_1 \le mR_m$ (r_1 is $\mathrm{mult}(x)/M^*$ for an x with a coordinate equal to 1), $r_\varphi \le R_m^2$. By (M1) when $N \ge d+1$: $r_0 \le \frac{1}{2}$ and $r_\varphi \le \max_{\ell \le d-2} 2^{\ell-d}/(\binom{\ell}{2}+1) = 1/(4(\binom{d-2}{2}+1))$ (the maximand is $1/g$ with N replaced by d).

T2.5 Lagrangian bound for the sharp cases. Fix the shape of the light member $Q - e_i$, with $t = |\mathrm{supp}Q|$ and $o = d-1-t$. Admissible pinned members have $b_i = 0$ and $P \subset [d] \setminus \mathrm{supp}Q$, a ground set of size $d-t$, so Lemma 41 gives $\sum s_2(P) \le \binom{d-t+1}{2} =: \mathrm{cap}$. The possible b fall into five classes:

class of b	count	coordinates available to P
$2e_j$, $j \in \mathrm{supp}Q$	t	$d-t$
$2e_j$, $j \notin \mathrm{supp}Q \cup \{i\}$	o	$d-t-1$
$e_j + e_l$, both in $\mathrm{supp}Q$	$\binom{t}{2}$	$d-t$
$e_j + e_l$, one in $\mathrm{supp}Q$	to	$d-t-1$
$e_j + e_l$, neither in $\mathrm{supp}Q$	$\binom{o}{2}$	$d-t-2$

Each b carries at most one pinned member. For every $\lambda \ge 0$ (weak duality),

$$\sum_{\text{pinned}} \theta_b \,\mathrm{mult}(P_b) \le \lambda\,\mathrm{cap} + \sum_b \max\Big(0,\ \max_P \big(\theta_b\,\mathrm{mult}(P) - \lambda s_2(P)\big)\Big),$$

where the inner maximum runs over partitions P of N with at most the available number of parts (s_2 depends only on the shape). Thus $\rho(F) \le \max(\rho_A, \rho_B)$, where ρ_A is this Lagrangian divided by W with no light member (all d A-classes with $d-1$ available coordinates and all $\binom{d}{2}$ B-classes with $d-2$, $\mathrm{cap} = \binom{d+1}{2}$), and $\rho_B = \max_Q[V_1(Q) + \mathrm{Lagrangian}(Q)]/W$ over partitions Q of $k+1$ into at most $d-1$ parts.

T2.6 Region reduction. Define (all from (4))

$$\begin{aligned}
E_1(d) &= \frac{8}{d} + \frac{d+1}{d-1}\,t_3(d) + \frac{2}{d-1}\Big(1+\frac{1}{d}\Big)\sqrt{12},\\
t_3(d) &= \max\Big\{\frac{1/3}{1-1/d}, \frac{1/6}{1-2/d}, \frac{2}{10}, \frac{8}{(\lfloor d/2\rfloor+1)(\lfloor d/2\rfloor+2)}\Big\},\\
E_2^u(d) &= \frac{8}{d} + \frac{d+1}{d-1}\cdot\frac{1}{\binom{d-2}{2}+1} + \frac{2}{d-1}\Big(1+\frac{1}{d}\Big)\sqrt{12},\\
E_2(d,m) &= \frac{2}{d}\frac{\min(1,R_m)}{\kappa_B^{(m)}} + \frac{d+1}{d-1}\frac{\min\big(\frac{1}{4(\binom{d-2}{2}+1)}, R_m^2\big)}{\kappa_B^{(m)}}\\
&\quad + \frac{\frac{2}{d-1}\min(1,R_m) + \frac{2}{d(d-1)}\min(1,mR_m)}{\sqrt{2\kappa_A^{(m)}\kappa_B^{(m)}}},
\end{aligned}$$

with $\kappa_B^{(m)} = (\frac{m}{m+1})^2$ and $\kappa_A^{(m)} = \frac{m^2}{(m+1)(m+2)}$.

- **Region I** ($k \le d-2$, $d \ge 40$). In (4) use $r_0, r_1 \le 1$ ($M^* = N!$), $\kappa_B \ge \frac{1}{4}$, $\kappa_A \ge \frac{1}{6}$ and, for the second term, $r_\varphi/\kappa_B \le 1/(\binom{N}{2}\max(1-\frac{k}{d}, \frac{1}{4})) \le t_3(d)$ (cases $k=1$, $k=2$, $3 \le k \le d/2$, $k > d/2$). So $S \le E_1(d)$, and every summand of E_1 is nonincreasing in d. $E_1(40) \le 0.741501$ [CA: `t2_interval/check_t2.py, stepC1`].

- **Region II** ($k \geq d-1$, $d \geq 20$; so $N \geq d+1$). (K) with $q_0 \geq 0$, (Moves) $r_\varphi \leq 1/(4(\binom{d-2}{2}+1))$, and $r_0, r_1 \leq 1$ give $S \leq E_2^u(d)$, nonincreasing in d. $E_2^u(20) \leq 0.790052$ [CA: `t2_interval/check_t2.py, stepC2`].

- **Region II′** ($3 \leq d \leq 19$, $k \geq (m_0(d)+1)d-2$, so $m = \lfloor N/d \rfloor \geq m_0(d)+1$ and $N \geq d+1$). Since $m = \lfloor (k+2)/d \rfloor \leq q_0 + 1$, (K) with $q_0 \geq m-1$ and (Moves) give $S \leq E_2(d,m)$. E_2 is nonincreasing in m: the $\kappa^{(m)}$ increase, and R_m, $mR_m = m!/m^{m-1}$, R_m^2 decrease. With $m_0 = 4, 3, 3$ for $d = 3, 4, 5$, $m_0 = 2$ for $6 \leq d \leq 16$ and $m_0 = 1$ for $17 \leq d \leq 19$ (minimal with $E_2(d, m_0) < 1$), $\max_d E_2(d, m_0) \leq 0.977398$ [CA: `t2_interval/check_t2.py, stepC3`].

- **Finite remainder** ($d \leq 39$, $k \leq d-2$; and $d \leq 19$, $d-1 \leq k \leq (m_0+1)d-3$): 1021 pairs. $S(d,k)$ is evaluated in interval arithmetic with $\mathrm{avg}_A, \mathrm{avg}_B$ exact rationals (closed form, checked against enumeration of C for $d \leq 7$). $S < 1$ except at 13 pairs; the largest value below 1 is ≤ 0.996011 at $(9,6)$ [CA: `t2_interval/check_t2.py, stepC4`].

- **The 13 sharp pairs** $(d,k) \in \{(3,1), (4,1), (4,2), (5,1), (5,2), (6,1), (6,2), (6,3), (7,1), (7,2), (7,3), (7,4), (8,5)\}$: T2.5 evaluated in interval arithmetic with stored multipliers λ (any $\lambda \geq 0$ is valid) gives $\rho \leq 0.679481$, attained at $(3,1)$ [CA: `t2_interval/check_t2.py, stepC5`].

By Lemma 43(1) the regions cover all $d \geq 3$, $k \geq 1$, and the supremum of the bounds is $\leq 0.996011 \leq 0.9961$. □

A.5 Part (d): $T = 3$, all $d \geq 3$, $k \geq 1$

Here $N = k+3$. The types of $|b| = 3$ are A ($3e_i$, $c = 1$, d of them), B ($2e_i + e_j$, $c = 3$, $d(d-1)$) and C ($e_i + e_j + e_l$, $c = 6$, $\binom{d}{3}$); in units of M^*, $\theta_X = \sqrt{c_X/\kappa_X}$ and $W = d\sqrt{\kappa_A} + d(d-1)\sqrt{3\kappa_B} + \binom{d}{3}\sqrt{6\kappa_C}$.

T3.1 Structure. By Lemma 39 (budget $n + n' - 4$) the members are of three kinds: **L** ($n = 1$): $Q - e_i$, $|Q| = k+1$, $Q_i = 0$; **M** ($n = 2$): $P - c$, $|c| = 2$, $|P| = k+2$, $\mathrm{supp}P \cap \mathrm{supp}c = \emptyset$; **Pin** ($n = 3$): $R - b$, $|R| = N$, $\mathrm{supp}R \cap \mathrm{supp}b = \emptyset$. L–L ($n + n' = 2$) and L–M ($n + n' = 3$) violate (S1), so a family has at most one L and never both an L and an M. M–M has budget 0: the P's are pairwise disjoint. Pin–Pin has budget 2: $\mathrm{ov}(R, R') \leq 2$ and the b's are distinct ($\mathrm{ov}(b,b) = 3$). L–Pin and M–Pin constraints are dropped. Hence $\rho(F) \leq (\max(V_L, V_M) + V_P)/W$, where V_L bounds one L, V_M all M's and V_P all pins.

T3.2 The three bounds (units of M^*; $r_{A2} = \binom{N}{2}\mathrm{Mb}(N-2, d-1)/M^*$, $r_1 = N\mathrm{Mb}(N-1, d-1)/M^*$, $r_0 = \mathrm{Mb}(N, d-1)/M^*$).

- V_L: the support of $Q - e_i$ is $\{e_i + c' : |c'| = 2\}$ with outputs $Q + c'$: $b = 3e_i$ gives $\mathrm{mult}(Q+2e_i) = \binom{N}{2}\mathrm{mult}(Q)$; $b = 2e_i + e_j$ gives $\mathrm{mult}(Q+e_i+e_j) = N\,\mathrm{mult}(Q+e_j)$; $b = e_i + 2e_j$ and $b = e_i + e_j + e_l$ give vectors of size N on $d-1$ coordinates. So $V_L = \theta_A r_{A2} + (d-1)\theta_B(r_1 + r_0) + \binom{d-1}{2}\theta_C r_0$.

- V_M: the support of $P - c$ is $\{c + e_j\}_{j \in [d]}$, so its value is

$$\begin{aligned} &\leq \theta_{\max} \sum_j \mathrm{mult}(P + e_j) = \theta_{\max} N\,\mathrm{mult}(P)\Big[(d-\ell) + \sum_{x \in \mathrm{supp}P} \frac{1}{P_x + 1}\Big] \\ &\leq \theta_{\max} N \mathrm{Mb}(N-1, \ell)(d - \ell/2) \end{aligned}$$

  with $\ell = |\mathrm{supp}P| \leq d-1$. The supports are disjoint, so $\sum \ell \leq d$ and $V_M = \theta_{\max}\, d\, N_f$ with $N_f = \max_{1 \leq \ell \leq d-1} N\mathrm{Mb}(N-1, \ell)(2d - \ell)/(2\ell M^*)$.

- V_P: at most d A-pins, each $\le \theta_A r_0$. For B- and C-pins, Lemma 41 ($r=3$) gives $\sum s_3(R) \le \binom{d+2}{3}$ and $\mathrm{mult}(R) \le \varphi_X s_3(R)$ with $\varphi_X = \max_{\ell \le \min(d-a_X, N)} \mathrm{Mb}(N,\ell)/s_3^{\min}(\ell,N)$, $a_B=2$, $a_C=3$. So $V_P = d\theta_A r_0 + \max(\theta_B\varphi_B, \theta_C\varphi_C)\binom{d+2}{3}/M^*$.

Define $S_3(d,k) = (\max(V_L,V_M)+V_P)/W$; then $\rho(F) \le S_3(d,k)$.

T3.3 Ratio lemmas. Let $q_0 = \lfloor k/d \rfloor$, $t = k/d$.

(K3) $\kappa_A \ge \frac{(q_0+1)^3}{(q_0+2)(q_0+3)(q_0+4)} \ge \frac{1}{24}$, $\kappa_B \ge \frac{(q_0+1)^3}{(q_0+2)^2(q_0+3)} \ge \frac{1}{12}$, $\kappa_C \ge (\frac{q_0+1}{q_0+2})^3 \ge \frac{1}{8}$: as (K), adding three units.

(K3-0) If $k \le d-3$ (g is 0/1, $M^*=N!$): $\kappa_C = \mathbb{E}2^{-X} \ge 1-\frac{3}{2}t$; $\kappa_B = \frac{1}{2}\mathbb{E}[3^{-g_1}2^{-g_2}] \ge \frac{1}{2}(1-\frac{7}{6}t)$; $\kappa_A = \mathbb{E}[1/(3+g_1)!] = \frac{1}{6}(1-\frac{3}{4}t)$.

(J3) If $N \ge 4$, then $\mathrm{Mb}(N,\ell)/s_3^{\min}(\ell,N) \le N!/\binom{N}{3}$ for all $\ell \le N$. *Proof.* Equality at $\ell = N$. At $\ell = N-1$: $\mathrm{Mb} = N!/2$, $s_3^{\min} = \binom{N-1}{3} + N - 2$, and $\binom{N}{3} \le 2\binom{N-1}{3} + 2(N-2) \iff \binom{N-1}{2} \le \binom{N-1}{3} + 2(N-2)$, true for $N \ge 6$ ($\binom{N-1}{3} \ge \binom{N-1}{2}$) and for $N = 4, 5$ ($3 \le 5$, $6 \le 10$). For $\ell \le N-2$: $\mathrm{Mb} \le N!2^{\ell-N}$, $s_3^{\min} \ge \binom{\ell}{3} + \ell$, and $h(\ell) = 2^{N-\ell}(\binom{\ell}{3}+\ell)$ is nonincreasing ($h(\ell)/h(\ell+1) \ge 1 \iff \binom{\ell}{3} + \ell - 1 \ge \binom{\ell}{2}$, true for $\ell \ge 1$); finally $h(N-2) - \binom{N}{3} = 4(\binom{n}{3}+n) - \binom{n+2}{3} = \frac{n}{2}(n^2-5n+10) \ge 0$ with $n = N-2$.

(J3′) If $N \ge d+1$: $\varphi_X/M^* \le \max_{\ell \le d-2} 2^{\ell-d}/(\binom{\ell}{3}+\ell) = 1/(4(\binom{d-2}{3}+d-2))$ by (M1) and the same monotonicity.

T3.4 Region reduction. Every step is either proved here or an evaluation by `t3_master/t3_bounds.py`.

- **U1** ($d \ge 100$, $1 \le k \le d-3$, so $N \le d$, $M^* = N!$). Use $W \ge \binom{d}{3}\sqrt{6\kappa_C}$; $r_{A2}, r_1, r_0 \le 1$; $N_f \le d/(N-1)$ (for $\ell \le N-1$, $N\mathrm{Mb}(N-1,\ell)/N! \le 2^{\ell+1-N}$ and $\ell 2^{N-1-\ell} \ge N-1$; for $\ell \ge N$ the ratio is $\le (2d-\ell)/(2\ell) \le d/N$); $\varphi_X/M^* \le 1/\binom{N}{3}$ (J3); and the larger of the (K3) and (K3-0) lower bounds for the κ's, with $\theta_{\max} = \max(1/\sqrt{\kappa_A}, \sqrt{3/\kappa_B}, \sqrt{6/\kappa_C})$. This gives

$$S_3 \le B_1(d,k) = \max\Big\{\frac{1}{\binom{d}{3}\sqrt{6\kappa_A\kappa_C}} + \frac{2(d-1)}{\binom{d}{3}\sqrt{2\kappa_B\kappa_C}} + \frac{3}{d}\frac{1}{\kappa_C},\ \frac{\theta_{\max} d^2}{(N-1)\binom{d}{3}\sqrt{6\kappa_C}}\Big\} + \frac{d}{\binom{d}{3}\sqrt{6\kappa_A\kappa_C}} + \max\Big\{\frac{1}{\sqrt{2\kappa_B\kappa_C}}, \frac{1}{\kappa_C}\Big\}\frac{\binom{d+2}{3}}{\binom{d}{3}\binom{N}{3}}.$$

  For fixed k, B_1 is decreasing in d. For $k < 12$ its value at $d = \max(100, k+3) = 100$ is at most 0.301833; for $k \ge 12$, with the crude (K3) constants, B_1 is also decreasing in N, so $B_1 \le B_1^{\mathrm{crude}}(100, N=15) \le 0.270687$ [CA: `t3_master/t3_bounds.py`].

- **U2** ($d \ge 100$, $N \ge d+1$). (K3) crude constants, $r_1, r_{A2} \le 1$, $r_0 \le \frac{1}{2}$ (M1), $N_f \le d/(d-1)$ (since $N\mathrm{Mb}(N-1,\ell) \le \mathrm{Mb}(N,\ell+1) \le M^* 2^{\ell+1-d}$ and $\ell 2^{d-1-\ell} \ge d-1$) and (J3′) give a bound $B_2(d)$ of the same form, decreasing in d, with $B_2(100) \le 0.128161$ [CA: `t3_master/t3_bounds.py`].

- **F** ($3 \le d \le 99$, $N \ge \max(m_0(d)d, d+1)$, $m = \lfloor N/d \rfloor$). (K3) with $q_0 \ge m-1$; (M2) with $t_0 = 0, 1, 2$: $r_0 \le \min(\frac{1}{2}, R_m)$, $r_1 \le \min(1, mR_m)$, $r_{A2} \le \min(1, m^2 R_m/2)$ ($m \ge 2$); $N\mathrm{Mb}(N-1,\ell) \le \mathrm{Mb}(N,\ell+1) \le M^* \min(2^{\ell+1-d}, R_m^{d-1-\ell})$; $\varphi_X/M^* \le \max_{\ell \le d-a_X} \min(2^{\ell-d}, R_m^{d-\ell})/(\binom{\ell}{3}+\ell)$. This gives $S_3 \le E_3(d,m)$, nonincreasing in m. The minimal m_0 with $E_3(d,m_0) < 1$ is 4

$(d = 3)$, 3 $(4 \le d \le 7)$, 2 $(8 \le d \le 14)$, 1 $(15 \le d \le 99)$, and $\max_d E_3(d, m_0) \le 0.968498$ [CA: `t3_master/t3_bounds.py`].

- **Finite** $(3 \le d \le 99,\ N < \max(m_0 d, d+1))$: 4774 pairs, $S_3(d,k)$ evaluated with exact rational κ's and interval square roots. $S_3 < 1$ except at 18 pairs; the largest value below 1 is ≤ 0.970559 at $(7,4)$ [CA: `t3_master/t3_bounds.py`].
- **The 18 residual pairs** $(d,k) \in \{(3,1), (4,1), (4,2), (5,1), (5,2), (6,1), (6,2), (6,3), (7,1), (7,2), (7,3), (8,1), (8,3), (8,4), (9,5), (10,6), (11,7), (12,8)\}$: each has an exact typed clique-LP certificate (Appendix C) with bound ≤ 0.7717523470 (attained at $(3,1)$) [CA: `t3_clique_lp/check_torus_lp.py;cover:t3_master/check_t3_cover.py`].

Monotonicity used above. In B_1 and B_2 every coefficient is a decreasing function of d: $\frac{2(d-1)}{\binom{d}{3}} = \frac{12}{d(d-2)}$, $\frac{d}{\binom{d}{3}}$, $\frac{d^2}{\binom{d}{3}} = \frac{6d}{(d-1)(d-2)}$, $\frac{d^2}{(d-1)\binom{d}{3}}$, $\frac{3}{d}$ and $\binom{d+2}{3}/\binom{d}{3} = \frac{(d+2)(d+1)}{d(d-1)}$, and $\binom{d+2}{3}/(\binom{d}{3}(\binom{d-2}{3}+d-2))$; the refined κ bounds of (K3-0) increase as $t = k/d$ decreases, so $\theta_{\max}$ and every $1/\sqrt{\kappa\kappa'}$ decrease; and $1/(N-1)$, $1/\binom{N}{3}$ decrease in N. In $E_3(d,m)$ the $\kappa^{(m)}$ increase with m (so W increases and the θ's decrease), while R_m, mR_m (ratio $(\frac{m}{m+1})^{m-1} \le 1$), $m^2 R_m/2$ for $m \ge 2$ (ratio $(\frac{m}{m+1})^{m-2} \le 1$; value 1 at $m = 2$, matching the value 1 used at $m = 1$) and $\min(2^{\ell-d}, R_m^{d-\ell})$ are nonincreasing; so E_3 is nonincreasing in m.

By Lemma 43(2) the regions cover all $d \ge 3$, $k \ge 1$, and the supremum of the bounds is $\le 0.970559 \le 0.9706$. □

A.6 Coverage of the (d,k) domain and exact comparison with 1

Lemma 43 (Coverage). *Let $m_0^{(2)}(d)$ and $m_0^{(3)}(d)$ be the tables of T2.6 and T3.4. (1) For $T = 2$, every pair (d,k) with $d \ge 3$, $k \ge 1$ lies in Region I, Region II, Region II′ or the finite remainder of T2.6, and the hypotheses used for the bound of that region hold there. (2) For $T = 3$, every pair with $d \ge 3$, $k \ge 1$ lies in U1, U2, F or the finite set of T3.4, and the hypotheses of that region hold there.*

Proof. (1) Fix $d \ge 3$ and let $N = k+2$, $q_0 = \lfloor k/d \rfloor$. If $d \ge 40$: $k \le d-2$ is Region I and $k \ge d-1$ is Region II. If $20 \le d \le 39$: $k \le d-2$ is finite and $k \ge d-1$ is Region II. If $3 \le d \le 19$: $k \le d-2$ and $d-1 \le k \le (m_0+1)d-3$ are finite, and $k \ge (m_0+1)d-2$ is Region II′. In each case the listed ranges partition $\{k \ge 1\}$, so there is no gap at any boundary. Hypotheses: Region I uses $k \le d-2$, i.e. $N \le d$ and $M^* = N!$. Region II uses $N \ge d+1$, which is $k \ge d-1$. Region II′ uses $N \ge d+1$ and $m := \lfloor N/d \rfloor \ge m_0+1$, both from $N \ge (m_0+1)d \ge 2d$; it also uses $q_0 \ge m-1$, which holds since $N = q_0 d + s + 2$ with $0 \le s \le d-1$ gives $N < (q_0+2)d$. Then (K) gives $\kappa_B \ge (\frac{q_0+1}{q_0+2})^2 \ge (\frac{m}{m+1})^2 = \kappa_B^{(m)}$ and $\kappa_A \ge \kappa_A^{(m)}$ ($x \mapsto x/(x+1)$ and $x \mapsto x^2/((x+1)(x+2))$ are increasing), so $S \le E_2(d,m) \le E_2(d,m_0)$ by the monotonicity in m proved in T2.6.

(2) Fix $d \ge 3$, $N = k+3$. If $d \ge 100$: $k \le d-3$ ($N \le d$) is U1 and $k \ge d-2$ ($N \ge d+1$) is U2. If $3 \le d \le 99$: $N < \max(m_0 d, d+1)$ is the finite set and $N \ge \max(m_0 d, d+1)$ is F. Again these partition $\{k \ge 1\}$. U1 uses $N \le d$ (so $M^* = N!$ and (K3-0) apply); for $k < 12$ the bound $B_1(d,k)$ is decreasing in d (the refined κ bounds increase as $t = k/d$ decreases, and every coefficient is decreasing in d), so it suffices to evaluate it at $d = \max(100, k+3) = 100$; for $k \ge 12$ the crude-constant bound is decreasing in d and in N, so it suffices to evaluate it at $(d,N) = (100,15)$. U2 uses $N \ge d+1$ for (M1) and (J3′). F uses $N \ge d+1$, $m = \lfloor N/d \rfloor \ge m_0$, and $q_0 \ge m-1$, which holds since $N = q_0 d + s + 3 < (q_0+2)d$ for $d \ge 3$; then $S_3 \le E_3(d,m) \le E_3(d,m_0)$. □

The checkers enumerate exactly these sets. `t2_interval/check_t2.py` evaluates $S(d,k)$ on `[(d,k) for d in range(3,40) for k in range(1,d-1)]` $\cup$ `[(d,k) for d in range(3,20) for k in range(d-1,(m0[d]+1)*d-2)]`, i.e. on $\{3 \le d \le 39,\ k \le d-2\} \cup \{3 \le d \le 19,\ d-1 \le k \le (m_0+1)d-3\}$, the finite remainder of Lemma 43(1) (1021 pairs). `t3_master/t3_bounds.py` evaluates $S_3(d,k)$ on `range(1, max(m0[d]*d, d+1) - 3)` for $3 \le d \le 99$, i.e. $1 \le k \le \max(m_0 d, d+1) - 4$, which is $N < \max(m_0 d, d+1)$ (4774 pairs). The tables $m_0^{(2)}$ ($4,3,3$ for $d = 3,4,5$; 2 for $6 \le d \le 16$; 1 for $17 \le d \le 19$) and $m_0^{(3)}$ (4 for $d=3$; 3 for $4 \le d \le 7$; 2 for $8 \le d \le 14$; 1 for $15 \le d \le 99$) are recomputed by the checkers as the least m with an interval upper bound < 1 and compared with the stored data. An independent audit script, `coverage_check.py` (shipped as a separate ancillary file next to `ca_appendix.zip`, so the archive hash is unchanged; SHA-256 `c42d11f652ed0bdd7daec55dd4bb9940319a8ff138c7f7d59956eacfa9ef0995`; it imports the shipped checkers unmodified and takes the path of the unzipped archive as its argument; re-run independently in QH-Lab with output identical to the original run), confirms on the box $3 \le d < 400$, $1 \le k < 1200$ that the enumerated sets equal the analytic definitions, that every pair lies in some region, and that the region hypotheses ($N \ge d+1$, $N \le d$, $m \ge m_0$, $q_0 \ge m-1$) hold pairwise; it also spot-checks the monotonicity claims (E_1, E_2^u in $d \le 2000$; E_2, E_3 in $m \le 60$, $m \le 40$; B_1, B_2 in d) and evaluates the master bounds directly on the boundary layers of every analytic region (208 pairs at $T=2$, max $S \le 0.45272$; 206 pairs at $T=3$, max $S_3 \le 0.33983$). These spot checks are consistency checks [Num]; the monotonicity statements themselves are proved in T2.6, T3.4 and above.

Exact comparison with 1; outward decimals. Every certified inequality is decided exactly. In `check_t2.py` and `t3_bounds.py` each bound is an `mpmath.iv` interval built from exact rationals (`fractions`), and the test is `v.b < 1` on the upper endpoint, a binary floating-point number compared exactly with 1. In `check_torus_lp.py` the bound is a `Fraction` (upper integer-square-root brackets in numerators, lower brackets in Λ_0), and the test `bound <= 1` is exact; the bookkeeping script `check_t3_cover.py` reads the printed LP bounds, which are rounded *up* (`math.ceil` at 10^{-10}). The decimals printed by `check_t2.py` and `t3_bounds.py` are round-to-nearest; the constants quoted in this appendix are instead outward (ceil) roundings of the interval upper endpoints: 0.741501, 0.790052, 0.977398, 0.996011, 0.679481 ($T=2$); 0.301833, 0.270687, 0.128161, 0.968498, 0.970559, 0.7717523470 ($T=3$). Their maxima give Theorem 13(c) $\rho \le 0.996011 \le 0.9961$ and (d) $\rho \le 0.970559 \le 0.9706$.

What is computer-assisted. In (c): the five evaluations C1–C5 (the 13 residual pairs at $T=2$ are closed by the interval Lagrangian bound C5, not by clique-LP certificates). In (d): the constants of U1, U2 and F, the 4774 interval evaluations, and the 18 clique-LP certificates. Everything else in this appendix, including (a), (b), the monotonicity statements and the validity of every bound evaluated, is proved above.

B Proof of Lemma 3

Primal and dual. With J the Choi operator of a channel $\mathrm{Sym}^T \to \mathrm{Sym}^{T+k}$, $P_K(T,k) = \max\{\mathrm{tr}(JX) : J \succeq 0,\ \mathrm{tr}_{\mathrm{out}}\, J = I\}$, where $X = \mathbb{E}_x v_x v_x^\dagger$ and $v_x = \bar\psi_x^{\otimes T} \otimes \psi_x^{\otimes(T+k)}$. The dual is $\min\{\mathrm{tr}\, Y : Y \otimes I \succeq X\}$. $J = I/D_{T+k}$ is strictly feasible, so strong duality holds and the dual optimum is attained (Slater).

Sector form of X. In the orthonormal occupation bases, $\psi_x^{\otimes m} = d^{-m/2} \sum_{|\gamma|=m} c_\gamma\, \omega^{x\,K\cdot\gamma} |\gamma\rangle$ with $c_\gamma = \sqrt{\mathrm{mult}(\gamma)}$ and $K \cdot \gamma = \sum_i \gamma_i k_i$. Hence $v_x = d^{-(2T+k)/2} \sum_{\alpha,\beta} c_\alpha c_\beta\, \omega^{x\,K\cdot(\beta-\alpha)} |\alpha,\beta\rangle$, and

$\mathbb{E}_x \omega^{x(u-u')} = [u \equiv u']$ gives $X = \sum_{s \in \mathbb{Z}_q} |w_s\rangle\langle w_s|$ with $w_s = d^{-(2T+k)/2} \sum_{K\cdot(\beta-\alpha)\equiv s} c_\alpha c_\beta |\alpha, \beta\rangle$.

Block structure of Y. Let $V_x = \bar{U}_x^{\otimes T} \otimes U_x^{\otimes (T+k)}$ with $U_x = \mathrm{diag}(\omega^{xk_j})$. Then $V_x X V_x^\dagger = X$, so if Y is feasible then so is $\bar{U}_x^{\otimes T} Y (\bar{U}_x^{\otimes T})^\dagger$, and so is their average $\bar{Y}$, with the same trace. $\bar{Y}$ commutes with every $\bar{U}_x^{\otimes T}$, which acts on $|\alpha\rangle$ by the phase $\omega^{-x\,K\cdot\alpha}$; hence $\bar{Y} = \bigoplus_w Y_w$ is block diagonal over the classes $A_w = \{\alpha : K \cdot \alpha \equiv w\}$. We assume this form.

Reduction to scalar constraints. The space splits into blocks $B_{w,\beta} = \mathrm{span}\{|\alpha, \beta\rangle : \alpha \in A_w\}$, on which $Y \otimes I$ acts as Y_w. The vector w_s meets block $B_{w,\beta}$ iff $K \cdot \beta - w \equiv s$, and there its component is $d^{-(2T+k)/2} c_\beta v_w$ with $v_w = \sum_{\alpha \in A_w} c_\alpha |\alpha\rangle$; each block meets exactly one w_s. So $Y \otimes I - X$ is a direct sum, over s, of $\bigoplus_{(w,\beta):K\cdot\beta - w \equiv s} Y_w$ minus the rank-one term $|w_s\rangle\langle w_s|$. For $D \succeq 0$ and a vector u, $D \succeq uu^\dagger$ iff $u \in \mathrm{range}(D)$ and $u^\dagger D^+ u \le 1$ (Schur complement). Writing $g_w = v_w^\dagger Y_w^+ v_w$ (and $g_w = +\infty$ if $v_w \notin \mathrm{range}(Y_w)$), and using $\sum_{\beta : K\cdot\beta \equiv u} c_\beta^2 = r_{T+k}(u)$ (each ordered $(T+k)$-tuple with sum u is counted once through its multiset), feasibility becomes

$$d^{-(2T+k)} \sum_w r_{T+k}(s+w)\, g_w \le 1 \qquad \forall s \in \mathbb{Z}_q.$$

Classes with $A_w = \emptyset$ do not occur, and for them $r_T(w) = 0$.

Objective. For $Y_w \succeq 0$ and $v_w \in \mathrm{range}(Y_w)$, Cauchy–Schwarz gives $\|v_w\|^4 = (v_w^\dagger Y_w^{1/2} Y_w^{+1/2} v_w)^2 \le (v_w^\dagger Y_w v_w)\, g_w \le \mathrm{tr}(Y_w) \|v_w\|^2 g_w$, so $\mathrm{tr}\, Y_w \ge \|v_w\|^2 / g_w = r_T(w)/g_w$, with equality for $Y_w = v_w v_w^\dagger / g_w$. Since g_w enters the constraints only through g_w itself, minimising $\mathrm{tr}\, Y$ is the same as minimising $\sum_w r_T(w)/z_w$ over $z_w = g_w > 0$ subject to the displayed constraints. This is Lemma 3. □

C Computer-assisted steps

Package and integrity. All computer-assisted (CA) steps are packaged as `ca_appendix.zip`, distributed as arXiv ancillary files and in the Zenodo record that accompanies this paper [33]. Its SHA-256 is

`e77e131bdd7becb0822ed582ccfc0a9e43b4b9183aeab37e65bec3e64d888b30`.

The archive contains eight claim folders, a shared checker (`common/check_torus_lp.py`), the scripts that generated the certificates (`generators/`), and a `SHA256SUMS` file with 32 entries. The coverage audit `coverage_check.py` (SHA-256 `c42d11f652ed0bdd7daec55dd4bb9940319a8ff138c7f7d59956eacfa9ef0995`) ships as a separate ancillary file next to the archive; see Appendix A.6. After unzipping, `sha256sum -c SHA256SUMS` verifies every file. Requirements: Python ≥ 3.10, `mpmath`, and `python-flint` 0.9.0. The clique-LP checker needs only the Python standard library.

Trust model. Every checker is standalone. It uses only exact rational arithmetic (`fractions`, with integer square roots for irrational quantities), interval arithmetic (`mpmath.iv`), or ball arithmetic (`python-flint arb`). No checker calls an LP, SDP or ODE solver, NumPy or SciPy. The certificates in `cert.json` and `*_data.json` were produced by floating-point generators (HiGHS for the clique LPs) in `generators/`. They are *data only*: weak duality makes any nonnegative multiplier valid, so correctness rests on the checkers alone. The package was re-run independently on a separate machine as part of QH-Lab [33] (v1.1.0, 49 automated tests), an independent Python re-implementation of the paper's tabulated values and worked examples. The archive hash and all 32 internal hashes matched, and every checker's output was identical to the shipped `expected_output.txt`. The three long clique-LP folders were run in resumable chunks, and their per-case lines were byte-identical.

The clique-LP checker (`check_torus_lp.py`). If two shifts Δ, Δ' both lie within ℓ_1-distance $T+k$ of a centre $c \in B_T$, then $\|\Delta - \Delta'\|_1 \le 2(T+k)$, which contradicts the separation condition (ii). Hence each such ball contains at most one family member, and for every $\mu \ge 0$ on centres,

$$\rho(F) = \sum_{\Delta \in F} r_\Delta \le \sum_c \mu_c + \sum_\Delta \max\Big(0,\, r_\Delta - \sum_{c:\|\Delta - c\|_1 \le T+k} \mu_c\Big), \qquad r_\Delta = \Lambda(\Delta)/\Lambda_0.$$

The certificate μ is a rational function of the centre's type. The checker brackets the weights a_b by integer square roots at scale 10^{30}, using upper brackets in numerators and lower brackets in Λ_0. It computes Z_b exactly via elementary symmetric polynomials, enumerates shift types independently of the generator, and compares the final bound with 1 exactly. Members with $n(\Delta) = 0$ cannot occur when $|F| \ge 2$ (Lemma S). The checker also verifies $r_\Delta < 1$ for unbalanced $\Delta \ge 0$. This is redundant given [9, Prop. X.5, Eq. (148)].

Folder-by-folder description

Runtimes are wall-clock times on a single CPU core and are indicative only.

`t2_interval` **(Theorem T2).** *Claim:* steps C1–C5 of the proof of Theorem T2 ($T = 2$, all $d \ge 3$, $k \ge 1$; $\rho(F) \le 0.9961$). *Method:* the claim splits into three parts.

- C1–C3 evaluate the closed-form region bounds $E_1(40)$, $E_2^u(20)$ and $E_2(d, m_0)$.
- C4 evaluates $S(d,k)$ in interval arithmetic on 1021 finite pairs, with exact rational $\mathrm{avg}_A, \mathrm{avg}_B$. $S < 1$ except at 13 pairs.
- C5 evaluates the Lagrangian bound at the 13 residual pairs with stored multipliers $\lambda \ge 0$.

Checker: `python3 check_t2.py` (reads `t2_data.json`; uses Fractions and `mpmath.iv`). *Output:* C1 0.741501, C2 0.790052, C3 max 0.977398, C4 $\max_{<1}$ 0.996011 at $(9,6)$, C5 max 0.679481; final line `THEOREM T2 COMPUTATIONAL STEPS: PASS ; overall sup bound <= 0.9960` (this line is printed round-to-nearest; the outward-rounded value is 0.996011, and the PASS test itself compares the interval upper endpoint with 1 exactly). *Runtime:* about 5 s.

`t3_master` **(Theorem T3, analytic part).** *Claim:* the constants of the region reduction for Theorem T3 ($T = 3$, all $d \ge 3$, $k \ge 1$; $\rho(F) \le 0.9706$). These are U1 (refined 0.301833, crude 0.2707), U2 (0.1282), and the m_0 table with $\max E_3 = 0.9685$. *Method:* the checker evaluates $S_3(d,k)$ on 4774 finite pairs with exact κ and interval square roots. $S_3 < 1$ except at 18 pairs, and the maximum below 1 is 0.97056 at $(7,4)$. A bookkeeping script then checks that the 18 residual pairs are covered by `t3_clique_lp`. *Checker:* `python3 t3_bounds.py 100 12 && python3 check_t3_cover.py`. The cover check reads `t3_bounds_DU100.json`. *Output:* `t3_bounds.py` ends with `analytic regions OK: True`; `check_t3_cover.py` prints `residual pairs: 18 ; max verified LP bound on them: 0.7717523470` and `THEOREM T3 (all d>=3, k>=1) COMPUTATIONAL STEPS: PASS`. *Runtime:* about 3 s.

`t3_clique_lp` **(Theorem T3, exact cases).** *Claim:* the packing inequality at $T = 3$ for $d = 2, \dots, 12$ with $k \le 20$ and for $d = 13, \dots, 20$ with $k \le 12$ (316 cases). This range includes the 18 residual pairs of `t3_master`. *Method:* typed clique-LP certificates with rational μ. *Checker:* `python3 check_torus_lp.py cert.json`. *Output:* `CASES=316 MAX_BOUND<=0.7717523470 ALL_PASS=True`. The maximum is attained at $(d,k) = (3,1)$, where the exact value is 0.77175235. *Runtime:* about 3 min.

`clique_lp_d16_Tk5` **(certified region).** *Claim:* the packing inequality for $d \le 16$ and $T + k \le 5$ (150 cases). *Checker:* as above. *Output:* `CASES=150 MAX_BOUND<=0.8403003465 ALL_PASS=True`. *Runtime:* about 3 s.

`d3_T60` **(certified region, $d = 3$).** *Claim:* the packing inequality for $d = 3$, $T \le 40$ with $k \le 3$, and $T = 41, \dots, 60$ with $k = 1$ (140 cases). *Checker:* as above. *Output:* `CASES=140 MAX_BOUND<=0.9991923171 A`
The maximum is at $T = 60$, consistent with $1 - \max \rho \approx 2.9/T^2$. *Runtime:* about 1 min.

`t45_clique_lp` **(evidence for Conjecture TT).** *Claim:* the packing inequality at $T = 4$ for $d = 3, \dots, 10$ with $k \le 12$ and $d = 11, \dots, 14$ with $k \le 10$, and at $T = 5$ for $d = 3, \dots, 9$ with $k \le 6$ (178 cases). *Checker:* as above. *Output:* `CASES=178 MAX_BOUND<=0.8866406634 ALL_PASS=True`. *Runtime:* about 2 min.

`su3_cert` **(SU(3) counterexample).** *Claim:* for $H = SU(3)$ acting on $V = \mathrm{Sym}^2\mathbb{C}^3$ and the Hessian group G (216 elements modulo phases), at $(T, k) = (1, 1)$ there is a G-orbit X whose imposed moment is matched exactly to that of the H-orbit, yet $P_X - P_H \ge 8.0399 \times 10^{-4}$, i.e. the finite orbit does not attain the continuous optimum. *Method:* ball arithmetic (`arb`, 256 bits). The checker certifies four things.

- Group closure by breadth-first search.
- Commutant dimension and projector traces.
- A Rayleigh lower bound on P_X and a congruence-plus-Gershgorin upper bound on P_H.
- An exactly moment-matched fiducial, found by Poincaré–Miranda on a preconditioned degree-≤ 4 map, with matching loss $\le 8.15 \times 10^{-10}$.

Checker: `python3 check_su3.py` (reads `su3_data.json`). *Output:* `gap_lower [0.000803995920422 +/- 1.42e-16] | matching_loss_upper [8.14587e-10 +/- 1.20e-17] | PASS True`. *Runtime:* about 4 s.

`spin1_cert` **(spin-1 counterexample).** *Claim:* the analogous strict gap for $SU(2)$ spin 1 with the octahedral group: an exactly moment-matched orbit with $P_X - P_H \ge 6.437 \times 10^{-4}$. *Method:* ball arithmetic. The checker certifies group closure and the character check $\sum \chi^2/24 = 3$, the quadrature weights, and a Cholesky factorisation (minimum pivot 1.76×10^{-12}). It then certifies $P_X \ge 0.516520734$ and $P_H \le 0.515877020$, together with an intermediate-value sign change of the matching function, with loss bound $72\delta/\|f\|$. *Checker:* `python3 check_spin1.py` (reads `spin1_data.json`). *Output:* `gap_lower [0.000643714380356 +/- 8.01e-17] | matching_loss_upper [7.20000e-12 +/- 1e-22] | PASS True`. *Runtime:* about 1 s.

File hashes

The six LP-folder copies of `check_torus_lp.py` are identical, including the copy in `common/`.

```
284885f682aa287943b2c356a01a61856ef8c19524c5491db3f65ca5530ef953  ./README.md
500fce4cabd10a639a457f56c28bb74d4ec3de8ae9c6b99f65b4d585c7a05da4  ./clique_lp_d16_Tk5/cert.json
abded76dbc9664403bef3f1d353e49dcf563c44d781bfcaee7c75ceab406b933  ./clique_lp_d16_Tk5/check_torus_lp.py
41de447e1afec21d4b7e453fe815417551b20043821da916d72afaaabd12b116  ./clique_lp_d16_Tk5/expected_output.txt
abded76dbc9664403bef3f1d353e49dcf563c44d781bfcaee7c75ceab406b933  ./common/check_torus_lp.py
4e1f176b9f97c453d60f773489a15b849edbd0c94d302b2bd68979da383527df  ./d3_T60/cert.json
abded76dbc9664403bef3f1d353e49dcf563c44d781bfcaee7c75ceab406b933  ./d3_T60/check_torus_lp.py
```

```
5fd48d2606e58d791a83493d76217d505f84d9a7c4f4d7eb0b1555ce16ee88d9  ./d3_T60/expected_output.txt
ee6920ce4bd8a0cd6edd90e544fce29ec03cfbbb9cba01cce9e47130afcf1197  ./generators/export_spin1.py
477f525cf30c19038bdbaf8cc8ff461a2dd512004a191b94b18699169f18ddb0  ./generators/export_su3.py
a48637ace29fbc91df98df07c256343b52d97923713159fbb62ca2e4ba825d27  ./generators/gen_torus_cert.py
b26f05d6859912e7be78c57b585e8a3a759bde9cceb864de784db722ef229d22  ./spin1_cert/check_spin1.py
98b6358ed32b875426b787b4d93b8c56bc248ec77006fd79a74411c6af232dc9  ./spin1_cert/expected_output.txt
496edcc41b5d150ee90c63fcc29b31b02be017b7d11143959f5749a0ca0a211c  ./spin1_cert/spin1_data.json
5cbce2c9ec3e80c90c817b18edcbfc6c49e7d76d40adb2dcba7de4585afaf75b  ./su3_cert/check_su3.py
2d3c3b46dc642108bbb9c5aba17085c691f83749b246e0ddc247b813dacd8ae5  ./su3_cert/expected_output.txt
54c1af221db10d21f91acd5f29a30ad47502fb7547e4a8c668b171c2ea1b0741  ./su3_cert/su3_data.json
7360fcb026a4cda4d0d39f2c7c1083518b8248c1c6caad9b0118b4436629e89e  ./t2_interval/check_t2.py
90ab1ee6f5472d41083fb15fce1771f9274145dbb72db688cd8735c75d40ccf4  ./t2_interval/expected_output.txt
4378b2d4ab43a82bade4c561fc1afc2723660e96ed0dc37ae6540b9dd9015481  ./t2_interval/make_t2_data.py
0912d57c75d71e7d445dea4670ef0dcfda87733cfd7d7c70fe55e4ae418045e0  ./t2_interval/t2_data.json
67dc7fbdf256efa51084eae5f6f3a742a625b2306783727b498701e360e27b0b  ./t3_clique_lp/cert.json
abded76dbc9664403bef3f1d353e49dcf563c44d781bfcaee7c75ceab406b933  ./t3_clique_lp/check_torus_lp.py
baefcd6d59d5e11c9d0551a8f6fe19e96bcb38bd06d7ac3dcbee89d560112272  ./t3_clique_lp/expected_output.txt
85399ef85a3e8cfeac493191cfedbd367fb5a70dd0641c3b0108bb2feab88566  ./t3_master/check_t3_cover.py
6d701bf17296510b8b30e84a34a0e12e99bebc307f4aa3f6515bd1ce59dc6859  ./t3_master/expected_cover_output.txt
00b9ff2e859e361127357a94639a7a1f038eea6919d69ace09613034976bb9d5  ./t3_master/expected_output.txt
3dcbb1beee39b34f72436241aacf60c426fe231dfb17b0caa74db7c3e6335235  ./t3_master/t3_bounds.py
59338e4eea8d368305114056617ff9ca07de8f006d1a2503cbb38f5c00c60b96  ./t3_master/t3_bounds_DU100.json
077f931afb2e786692ab2733cff53c556f4615e5e33c2d16c998973d446b7d95  ./t45_clique_lp/cert.json
abded76dbc9664403bef3f1d353e49dcf563c44d781bfcaee7c75ceab406b933  ./t45_clique_lp/check_torus_lp.py
6278e682a9d6a6e9d2c71121d242fa7aead8ee6e1c98f964ec32e1a5281a3506  ./t45_clique_lp/expected_output.txt
```